

Exercises**Level - 1****(Problems Based on Fundamentals)**

1. Solve for θ : $\sin 3\theta = 0$
2. Solve for θ : $\cos^2 (5\theta) = 0$
3. Solve for θ : $\tan \theta = \sqrt{3}$
4. Solve for θ : $\sin 2\theta = \sin \theta$
5. Solve for θ : $\sin (9\theta) = \sin \theta$
6. Solve for θ : $5\sin^2 \theta + 3\cos^2 \theta = 4$
7. Solve for θ : $\tan(\theta - 15^\circ) = 3 \tan (\theta + 15^\circ)$
8. Solve for θ : $\tan^2(\theta) + \cot^2(\theta) = 2$
9. Solve for θ : $\cos (\theta) + \cos (2\theta) + \cos (3\theta) = 0$
10. Solve for θ : $\sin (2\theta) + \sin (4\theta) + \sin (6\theta) + \tan (2\theta) = 1$
11. Solve for θ : $\tan (\theta) + \tan (2\theta) + \tan (\theta) + \tan (2\theta) = 1$
12. Solve for θ : $\tan (\theta) + \tan (2\theta) + \tan(3\theta) = \tan (\theta) \cdot \tan (\theta)$
13. Solve for θ : $\cot^2 \theta + \frac{3}{\sin \theta} + 3 = 0$
14. Solve for θ : $2 \tan \theta - \cot \theta = -1$

15. Solve for θ : $\tan^2 \theta + (1 - \sqrt{3}) \tan \theta - \sqrt{3} = 0$
 16. Solve for θ : $\tan \theta + \tan \left(\theta + \frac{\pi}{3} \right) + \tan \left(\theta + \frac{2\pi}{3} \right) = 3$
 17. Solve for θ : $3 \tan (\theta - 60^\circ) = \tan (\theta + 60^\circ)$
 18. Solve for θ : $\tan \theta + \tan 2\theta + \tan 3\theta = 0$
 19. Solve for θ : $\cos 2\theta \cos 4\theta = \frac{1}{2}$
 20. Solve for θ : $\cot \theta - \tan \theta = \cos \theta - \sin \theta$
 21. Solve for θ : $(1 - \tan \theta)(1 + \sin 2\theta) = 1 + \tan \theta$
 22. Solve for θ : $2 \sin^2 \theta + \sin^2 2\theta = 2$
 23. Solve for θ : $\sin 3\alpha = 4 \sin \alpha \sin (\theta + \alpha) \sin (\theta - \alpha)$,
 $\alpha \neq n\pi, n \in \mathbb{Z}$
 24. Solve for θ : $4 \sin \theta \sin 2\theta \sin 4\theta = \sin 3\theta$
- A trigonometric equation is of the form
 $a \cos \theta \pm b \sin \theta = c$
25. Solve for θ : $\sin (\theta) + \cos (\theta) = 1$
 26. Solve for θ : $\sqrt{3} \sin (\theta) + \cos (\theta) = 2$
 27. Solve for θ :
 $\sin (2\theta) + \cos (2\theta) = \sin (\theta) + \cos (\theta) + 1 = 0$
 28. Solve for θ : $\sin^3 \theta + \sin \theta \cos \theta + \cos^3 \theta = 1$
 29. Solve for θ : $\sin \theta + \sqrt{3} \cos \theta = \sqrt{2}$
 30. Solve for θ : $\sqrt{2} \sec \theta + \tan \theta = 1$
 31. Solve for θ : $\cot \theta + \operatorname{cosec} \theta = \sqrt{3}$
 32. Solve for θ : $\sin \theta + \cos \theta = \sqrt{2}$
 33. Solve for θ : $\sqrt{3} \cos \theta + \sin \theta = 1$
 34. Solve for θ : $\sin \theta + \cos \theta = 1$
 35. Solve for θ : $\operatorname{cosec} \theta = 1 + \cot \theta$
 36. Solve for θ : $\tan \theta + \sec \theta = \sqrt{3}$
 37. Solve for θ : $\cos \theta + \sqrt{3} \sin \theta = 2 \cos 2\theta$
 38. Solve for θ :
 $\sqrt{3}(\cos \theta - \sqrt{3} \sin \theta) = 4 \sin 2\theta \cdot \cos 3\theta$

PRINCIPAL VALUE

39. Find the principal value of $\sin (\theta) = -\frac{1}{2}$.
40. Find the principal value of $\sin (\theta) = \frac{1}{\sqrt{2}}$
41. Find the principal value of $\tan (\theta) = -\sqrt{3}$
42. Find the principal values of $\tan \theta = -1$
43. Find the principal values of $\cos \theta = \frac{1}{2}$
44. Find the principal values of $\cos \theta = -\frac{1}{2}$
45. Find the principal values of $\tan \theta = -\sqrt{3}$
46. Find the principal values of $\sec \theta = \sqrt{2}$.

Solutions in case if two equations are given:

47. If $\sin (\theta) = \frac{1}{\sqrt{2}}$ and $\cos (\theta) = -\frac{1}{\sqrt{2}}$,
then find the general values of θ
48. If $\sin (\theta) = \frac{1}{\sqrt{2}}$ and $\tan (\theta) = -1$, then find the general values of θ
49. If $\cos \theta = \frac{1}{\sqrt{2}}$ and $\tan \theta = -1$, then find the general value of θ
50. Find the most general value of θ which satisfy the equations $\sin \theta = \frac{1}{2}$ and $\tan \theta = \frac{1}{\sqrt{3}}$
51. If $(1 + \tan A)(1 + \tan B) = 2$, then find all the values of $A + B$
52. If $\tan (A - B) = 1$ and $\sec(A + B) = \frac{2}{\sqrt{3}}$, then find the smallest +ve values of A and B and their most general values.
53. If $\sin (\pi \cos \theta) = \cos (\pi \sin \theta)$, then prove that,
 $\cos \left(\theta \pm \frac{\pi}{4} \right) = \frac{1}{2\sqrt{2}}$.
54. If $\tan (\pi \cos \theta) = \cot (\pi \sin \theta)$, then prove that
 $\cos \left(\theta - \frac{\pi}{4} \right) = \frac{1}{2\sqrt{2}}$
55. If $\sin A = \sin B$ and $\cos A = \cos B$, then find the values of A in terms of B .
56. If A and B are acute +ve angles satisfying the equations
 $3 \sin^2 A + 2 \sin^2 B = 1$ and $3 \sin^2 A - 2 \sin^2 B = 0$, then find $A + 2B$.
57. Solve $x + y = \frac{\pi}{4}$ and $\tan x + \tan y = 1$.
58. Solve $\sin x + \sin y = 1$, $\cos 2x - \cos 2y = 1$.
59. Solve $x + y = \frac{2\pi}{3}$ and $\sin x = 2 \sin y$.
60. Solve $x + y = \frac{2\pi}{3}$ and $\cos x + \cos y = \frac{3}{2}$
61. If $r \sin \theta = 3$ and $r = 4(1 + \sin \theta)$, where $0 \leq \theta \leq 2\pi$, then find the value of θ
62. Solve $\sin x + \sin y = 1$
 $\cos 2x - \cos 2y = 1$
63. Find the co-ordinates of the point of intersection of the curves $y = \cos x$ and $y = \sin 2x$
64. Find all points of x, y that satisfying the equations
 $\cos x + \cos y + \cos (x + y) = -\frac{3}{2}$
65. If $0 < \theta, \phi < \pi$ and $8 \cos \theta \cos \phi \cos (\theta + \phi) + 1 = 0$, then find θ and ϕ .

66. Find the set of values of x for which $\frac{\tan 3x - \tan 2x}{1 + \tan 3x \cdot \tan 2x} = 1$
67. Find the number of solutions of the equation $\tan x + \sec x = 2 \cos x$ lying in the interval $[0, 2\pi]$
68. Find the number of values of x in the interval $[0, 3\pi]$ satisfying the equation $2 \sin^2 x + 5 \sin x - 3 = 0$
69. Find the smallest positive value of x such that $\tan(x + 20^\circ) = \tan(x - 10^\circ) \cdot \tan x \cdot \tan(x + 10^\circ)$
70. If $\sin^2 x + \cos^2 y = 2 \sec^2 z$, then find x, y and z .

DIFFERENT TYPES OF TRIGONOMETRIC EQUATIONS

Type 1

71. Solve $5 \cos 2x + 2 \cos^2\left(\frac{x}{2}\right) + 1 = 0$
72. Solve $4 \sin^4 x + \cos^4 x = 1$
73. Solve $4 \cos^2 x \sin x - 2 \sin^2 x = 2 \sin x$
74. Solve $\sin 3x + \cos 2x = 1$
75. Solve $2 \cos 2x + \sqrt{2} \sin x = 2$
76. Solve $1 + \sin^2 x + \cos^2 x = \frac{3}{2} \sin 2x$
77. Solve $\sin^6 x + \cos^6 x = \frac{7}{16}$
78. Solve $\sin^8 x + \cos^8 x = \frac{17}{16} \cos^2 2x$
79. Solve $2 \sin^3 x + 2 = \cos^2 3x$
80. Solve $\cos 4x = \cos^2 3x$
81. Solve $\cos 2x = 6 \tan^2 x - 2 \cos^2 x$

Type 2

82. Solve $(2 \sin x - \cos x)(1 + \cos x) = \sin^2 x$
83. Solve $2 \sin^2 x + \sin x - 1 = 0$
where $0 \leq x \leq 2\pi$
84. Solve $5 \sin^2 x + 7 \sin x - 6 = 0$,
where $0 \leq x \leq 2\pi$
85. Solve $\sin^2 x - \cos x = \frac{1}{4}$, where $0 \leq x \leq 2\pi$
86. Solve $\tan^2 x - 2 \tan x - 3 = 0$
87. Solve $2 \cos^2 x - \sqrt{3} \sin x + 1 = 0$

Type 3

88. Solve $\sin x + \sin 3x + \sin 5x = 0$, $0 \leq x \leq \frac{\pi}{2}$
89. Solve $\cos x - \cos 2x = \sin 3x$
90. Solve $\sin 7x + \sin 4x + \sin x = 0$, $0 \leq x \leq \frac{\pi}{2}$
91. Solve $\cos 3x + \cos 2x$
 $= \sin\left(\frac{3x}{2}\right) + \sin\left(\frac{x}{2}\right)$, $0 \leq x \leq 2\pi$

92. Solve $\sin x + \sin 2x + \sin 3x$
 $= \cos x + \cos 2x + \cos 3x$, $-\pi \leq x \leq 2\pi$
93. Solve $\cos 2x + \cos 4x = 2 \cos x$
94. Solve $\sin 2x + \cos 2x + \sin x + \cos x + 1 = 0$
95. Solve $\tan x + \tan 2x + \tan 3x = 0$
96. Solve $\tan 3x + \tan x = 2 \tan 2x$
97. Solve $(1 - \tan x)(1 + \sin 2x) = (1 + \tan x)$
98. Solve $\sin x - 3 \sin 2x + \sin 3x$
 $= \cos x \cdot 3 \cos 2x + \cos 3x$

Type 4

99. Solve $4 \sin x \cdot \sin 2x \cdot \sin 4x = \sin 3x$
100. Solve $\cos x \cdot \cos 2x \cdot \cos 3x = 1/4$, $0 \leq x \leq 2\pi$
101. Solve $\sin 3\alpha = 4 \sin \alpha \cdot \sin(x + \alpha) \cdot \sin(x - \alpha)$
102. Solve $\sin 2x \cdot \sin 4x + \cos 2x = \cos 6x$
103. Solve $\sec x \cdot \cos 5x + 1 = 0$, $0 \leq x \leq 2\pi$
105. Solve $\cos x \cdot \cos 6x = -1$

Type 5

106. Solve $2 \sin^2 x - 5 \sin x \cos x - 8 \cos^2 x = -2$
107. Solve $5 \sin^2 x - 7 \sin x \cos x + 10 \cos^2 x = 4$
108. Solve $2 \sin^2 x - 5 \sin x \cos x - 8 \cos^2 x = -3$
109. Solve $\sin^3 x \cos x + \sin^2 x \cos^2 x + \sin x \cos^3 x = 1$

Type 6

110. Solve $1 + 2 \operatorname{cosec} x = -\frac{\sec^2\left(\frac{x}{2}\right)}{2}$
111. Solve $(\cos x - \sin x)(2 \tan x + \sec x) + 2 = 0$
112. Solve $\frac{\sin^3 \frac{x}{2} - \cos^3 \frac{x}{2}}{2 + \sin x} = \frac{\cos x}{3}$
113. Solve $\cot\left(\frac{x}{2}\right) - \operatorname{cosec}\left(\frac{x}{2}\right) = \cot x$
114. If $\theta_1, \theta_2, \theta_3, \theta_4$ be the four roots of the equation $\sin(\theta + \alpha) = k \sin 2\theta$, no two of which differ by a multiple of 2π , then prove that
 $\theta_1 + \theta_2 + \theta_3 + \theta_4 = (2n + 1)\pi, n \in \mathbb{Z}$

Type 7

115. Solve $\sin x + \cos x - 2\sqrt{2} \sin x \cos x = 0$
116. Solve $\sin^3 x + \sin x \cos x + \cos^3 x = 1$
117. Solve $\sin x + \cos x = 1 - \sin x \cos x$
118. Solve $1 + \sin^3 x + \cos^3 x = \frac{3}{2} \sin 2x$
119. Solve $\sin 2x - 12(\sin x - \cos x) + 12 = 0$, where $0 \leq x \leq 2\pi$
120. Solve $\sin 6x + \cos 4x + 2 = 0$

Type 8

121. Solve $\sin^6 x = 1 + \cos^4 3x$
122. Solve $\sin^4 x = 1 + \tan^8 x$
123. Solve $\sin^2 x + \cos^2 y = 2 \sec^2 z$

124. Solve $\sin 3x + \cos 2x + 2 = 0$

125. Solve $\cos 4x + \sin 5x = 2$

Type 9126. Find the values of x in $(-\pi, \pi)$ which satisfy the equation

$$8^{1+|\cos x|+\cos^2 x+|\cos x|^3+\cos^4 x+|\cos x|^5+\dots\text{to}\infty} = 64$$

127. Solve $2^{1+|\cos x|+\cos^2 x+|\cos x|^3+\cos^4 x+|\cos x|^5+\dots\text{to}\infty} = 4$

128. Solve $1 + \sin \theta + \sin^2 \theta + \sin^3 \theta + \dots \text{to } \infty$

$$= 4 + 2\sqrt{3}$$

129. Solve $|\cos x|^{\sin^2 x - \frac{3}{2}\sin x + \frac{1}{2}} = 1$

130. Solve $e^{\sin x} - e^{-\sin x} - 4 = 0$

131. If $e^{[\sin^2 x + \sin^4 x + \sin^6 x + \dots \text{to } \infty] \log_e 2}$ satisfies the equations $x^2 - 9x + 8 = 0$, then find the value of

$$\frac{\cos x}{\cos x + \sin x}, 0 < x < \frac{\pi}{2}$$

132. Solve $\log_{\cos x} \tan x + \log_{\sin x} \cot x = 0$

133. Solve $3^{\sin 2x + 2\cos^2 x} + 3^{1 - \sin 2x + 2\sin^2 x} = 28$

134. Solve $\log_{\cos x} \sin x + \log_{\sin x} \cos x = 2$, where $x > 0$

Type 10

135. Solve $2 \cos^2 \left(\frac{x}{2} \right) \sin^2 x = x^2 + \frac{1}{x^2}, 0 < x < \frac{\pi}{2}$

136. Solve $2 \cos^2 \left(\frac{x^2 + x}{6} \right) = 2^x + 2^{-x}$

HINTS AND SOLUTIONS

LEVEL 1

1. We have,
- $\sin 3\theta = 0$

$$\Rightarrow 3\theta = n\pi$$

$$\Rightarrow \theta = \frac{n\pi}{3}, \text{ where } n \in I$$

2. We have,
- $\cos^2(5\theta) = 0$

$$\Rightarrow \cos^2(5\theta) = \cos^2\left(\frac{\pi}{2}\right)$$

$$\Rightarrow (5\theta) = n\pi \pm \left(\frac{\pi}{2}\right)$$

$$\Rightarrow \theta = \frac{1}{5}\left(n\pi \pm \left(\frac{\pi}{2}\right)\right), \text{ where } n \in I$$

3. We have,
- $\tan \theta = \sqrt{3}$

$$\Rightarrow \tan \theta = \tan\left(\frac{\pi}{3}\right)$$

$$\Rightarrow \theta = n\pi + \left(\frac{\pi}{3}\right), \text{ where } n \in I$$

4. We have,
- $\sin 2\theta = \sin \theta$

$$\Rightarrow 2 \sin \theta \cos \theta = \sin \theta$$

$$\Rightarrow \sin \theta (2 \cos \theta - 1) = 0$$

$$\Rightarrow \sin \theta = 0 \text{ and } (2 \cos \theta - 1) = 0$$

$$\Rightarrow \sin \theta = 0 \text{ and } \cos \theta = \frac{1}{2}$$

$$\Rightarrow \theta = n\pi \text{ and } \theta = 2n\pi \pm \frac{\pi}{3}, \text{ where } n \in I$$

5. We have,
- $\sin(9\theta) = \sin \theta$

$$\Rightarrow \sin(9\theta) - \sin \theta = 0$$

$$\Rightarrow 2 \cos\left(\frac{9\theta + \theta}{2}\right) \sin\left(\frac{9\theta - \theta}{2}\right) = 0$$

$$\Rightarrow 2 \cos(5\theta) \sin(4\theta) = 0$$

$$\Rightarrow \cos(5\theta) = 0 \text{ and } \sin(4\theta) = 0$$

$$\Rightarrow (5\theta) = (2n+1)\frac{\pi}{2} \text{ and } (4\theta) = n\pi$$

$$\Rightarrow \theta = (2n+1)\frac{\pi}{10} \text{ and } \theta = \left(\frac{n\pi}{4}\right)$$

where $n \in I$

6. We have,
- $5 \sin^2 \theta + 3 \cos^2 \theta = 4$

$$\Rightarrow 2 \sin^2 \theta + 3(\sin^2 \theta + \cos^2 \theta) = 4$$

$$\Rightarrow 2 \sin^2 \theta + 3 = 4$$

$$\Rightarrow 2 \sin^2 \theta = 1$$

$$\Rightarrow \sin^2 \theta = \frac{1}{2}$$

$$\Rightarrow \sin^2 \theta = \left(\frac{1}{\sqrt{2}}\right)^2 = \sin^2\left(\frac{\pi}{4}\right)$$

$$\Rightarrow \theta = n\pi \pm \left(\frac{\pi}{4}\right), \text{ where } n \in I$$

7. We have,
- $\tan(\theta - 15^\circ) = \tan(\theta + 15^\circ)$

$$\Rightarrow \frac{\tan(\theta - 15^\circ)}{\tan(\theta + 15^\circ)} = \frac{3}{1}$$

$$\Rightarrow \frac{\tan(\theta - 15^\circ) + \tan(\theta + 15^\circ)}{\tan(\theta - 15^\circ) - \tan(\theta + 15^\circ)} = \frac{3+1}{3-1}$$

$$\Rightarrow \frac{\sin(\theta + 15^\circ + \theta - 15^\circ)}{\sin(\theta + 15^\circ - \theta + 15^\circ)} = \frac{3+1}{3-1}$$

$$\Rightarrow 2 \sin(2\theta) = 2$$

$$\Rightarrow \sin(2\theta) = 1$$

$$\Rightarrow \theta = (4n+1)\frac{\pi}{4}, n \in I$$

8. We have,
- $\tan^2(\theta) + \cot^2(\theta) = 2$

$$\Rightarrow \tan^2(\theta) + \frac{1}{\tan^2(\theta)} = 2$$

$$\Rightarrow \tan^4(\theta) - 2 \tan^2(\theta) + 1 = 0$$

$$\Rightarrow (\tan^2(\theta) - 1)^2 = 0$$

$$\Rightarrow (\tan^2(\theta) - 1) = 0$$

$$\Rightarrow \theta = n\pi \pm \left(\frac{\pi}{4}\right), n \in I$$

9. We have,
- $(\cos(3\theta) + \cos(\theta)) + \cos(2\theta) = 0$

$$\Rightarrow (\cos(3\theta) + \cos(\theta)) + \cos(2\theta) = 0$$

$$\Rightarrow 2 \cos(2\theta) \cos(\theta) + \cos(2\theta) = 0$$

$$\Rightarrow \cos(2\theta) (2 \cos(\theta) + 1) = 0$$

$$\Rightarrow \cos(2\theta) = 0 \text{ and } (2 \cos(\theta) + 1) = 0$$

$$\Rightarrow \cos(2\theta) = 0 \text{ and } \cos(\theta) = -\frac{1}{2}$$

$$\Rightarrow \theta = (2n+1)\left(\frac{\pi}{4}\right) \text{ and } \theta = n\pi \pm \left(\frac{2\pi}{3}\right)$$

$$\begin{aligned} 10. \text{ We have } \sin(2\theta) + \sin(4\theta) + \sin(6\theta) &= 0 \\ \sin(6\theta) + \sin(2\theta) + \sin(4\theta) &= 0 \\ 2\sin(4\theta), \cos(2\theta) + \sin(4\theta) &= 0 \\ \sin(4\theta)(2\cos(2\theta) + 1) &= 0 \\ \sin(4\theta) = 0 \text{ and } (2\cos(2\theta) + 1) &= 0 \end{aligned}$$

$$\Rightarrow (4\theta) = n\pi \text{ and } \cos(2\theta) = -\frac{1}{2}$$

$$\Rightarrow \theta = \left(\frac{n\pi}{4}\right) \text{ and } (2\theta) = n\pi \pm \left(\frac{2\pi}{3}\right)$$

$$\Rightarrow \theta = \left(\frac{n\pi}{4}\right) \text{ and } \theta = \left(\frac{n\pi}{2}\right) \pm \left(\frac{\pi}{3}\right), n \in I$$

11. We have

$$\begin{aligned} \tan(\theta) + \tan(2\theta) + \tan(\theta)\tan(2\theta) &= 1 \\ \Rightarrow \tan(2\theta) + \tan(\theta) &= 1 - \tan(\theta)\tan(2\theta) \end{aligned}$$

$$\Rightarrow \frac{\tan(2\theta) + \tan(\theta)}{1 - \tan(\theta)\tan(2\theta)} = 1$$

$$\Rightarrow \tan(3\theta) = 1$$

$$\Rightarrow \tan(3\theta) = \tan\left(\frac{\pi}{4}\right)$$

$$\Rightarrow (3\theta) = n\pi + \left(\frac{\pi}{4}\right)$$

$$\Rightarrow \theta = \left(\frac{n\pi}{3}\right) + \left(\frac{\pi}{12}\right), n \in I$$

12. We have

$$\begin{aligned} \tan(\theta) + \tan(2\theta) + \tan(3\theta) \\ = \tan(\theta), \tan(2\theta), \tan(3\theta) \end{aligned}$$

$$\Rightarrow \tan(\theta) + \tan(2\theta) = -\tan(3\theta) + \tan(\theta)\tan(2\theta)\tan(3\theta)$$

$$\Rightarrow \tan(\theta) + \tan(2\theta) = -\tan(3\theta)(1 - \tan(\theta)\tan(2\theta))$$

$$\Rightarrow \left(\frac{\tan(\theta) + \tan(2\theta)}{(1 - \tan(\theta)\tan(2\theta))}\right) = -\tan(3\theta)$$

$$\Rightarrow \tan(3\theta) = -\tan(3\theta)$$

$$\Rightarrow 2\tan(3\theta) = 0$$

$$\Rightarrow (3\theta) = n\pi$$

$$\Rightarrow \theta = \left(\frac{n\pi}{3}\right), n \in I$$

13. Given equation is

$$\cot^2\theta + \frac{3}{\sin\theta} + 3 = 0$$

$$\Rightarrow \cot^2\theta + 3(1 + \operatorname{cosec}\theta) = 0$$

$$\Rightarrow (\operatorname{cosec}^2\theta - 1) + 3(1 + \operatorname{cosec}\theta) = 0$$

$$\Rightarrow (\operatorname{cosec}\theta - 1 + 3)(1 + \operatorname{cosec}\theta) = 0$$

$$\Rightarrow (\operatorname{cosec}\theta + 2)(1 + \operatorname{cosec}\theta) = 0$$

$$\Rightarrow \operatorname{cosec}\theta = -1, -2$$

$$\Rightarrow \sin\theta = -1, \sin\theta = -\frac{1}{2}$$

$$\Rightarrow \theta = (4n-1)\frac{\pi}{2}, \theta = n\pi + (-1)^n\left(-\frac{\pi}{6}\right), n \in I$$

14. Given equation is

$$2\tan\theta - \cot\theta = -1$$

$$\Rightarrow 2\tan\theta = \cot\theta - 1$$

$$\Rightarrow 2\tan\theta = \frac{1}{\tan\theta} - 1$$

$$\Rightarrow 2\tan^2\theta + \tan\theta - 1 = 0$$

$$\Rightarrow 2\tan^2\theta + 2\tan\theta - \tan\theta - 1 = 0$$

$$\Rightarrow 2\tan\theta(\tan\theta + 1) - (\tan\theta + 1) = 0$$

$$\Rightarrow (2\tan\theta - 1)(\tan\theta + 1) = 0$$

$$\Rightarrow (2\tan\theta - 1) = 0, (\tan\theta + 1) = 0$$

$$\Rightarrow \tan\theta = -1, \frac{1}{2}$$

$$\Rightarrow \theta = \left(n\pi - \frac{\pi}{4}\right), \theta = n\pi + \alpha, \alpha = \tan^{-1}\left(\frac{1}{2}\right)$$

15. Given equation is

$$\tan^2\theta + (1 - \sqrt{3})\tan\theta - \sqrt{3} = 0$$

$$\Rightarrow \tan^2\theta + \tan\theta - \sqrt{3}(\tan\theta + 1) = 0$$

$$\Rightarrow \tan\theta(\tan\theta + 1) - \sqrt{3}(\tan\theta + 1) = 0$$

$$\Rightarrow (\tan\theta - \sqrt{3})(\tan\theta + 1) = 0$$

$$\Rightarrow \tan\theta = \sqrt{3}, \tan\theta = -1$$

$$\Rightarrow \theta = n\pi + \frac{\pi}{3}, \theta = n\pi - \frac{\pi}{4}, n \in I$$

16. Given equation is

$$\tan\theta + \tan\left(\theta + \frac{\pi}{3}\right) + \tan\left(\theta + \frac{2\pi}{3}\right) = 3$$

$$\Rightarrow \tan\theta + \tan\left(\frac{\pi}{3} + \theta\right) + \tan\left(\pi - \left(\frac{\pi}{3} - \theta\right)\right) = 3$$

$$\Rightarrow \tan\theta + \tan\left(\frac{\pi}{3} + \theta\right) - \tan\left(\frac{\pi}{3} - \theta\right) = 3$$

$$\Rightarrow \tan\theta + \frac{\sqrt{3} + \tan\theta}{1 - \sqrt{3}\tan\theta} - \frac{\sqrt{3} - \tan\theta}{1 + \sqrt{3}\tan\theta} = 3$$

$$\Rightarrow \tan\theta + \frac{8\tan\theta}{1 - 3\tan^2\theta} = 3$$

$$\Rightarrow \frac{9\tan\theta - 3\tan^3\theta}{1 - 3\tan^2\theta} = 3$$

$$\Rightarrow \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} = 1$$

$$\Rightarrow \tan(3\theta) = 1$$

$$\Rightarrow 3\theta = n\pi + \frac{\pi}{4}$$

$$\Rightarrow \theta = \frac{n\pi}{3} + \frac{\pi}{12}, n \in I$$

17. Given equation is

$$3 \tan(\theta - 60^\circ) = \tan(\theta + 60^\circ)$$

$$\Rightarrow 3 = \frac{\tan(\theta + 60^\circ)}{\tan(\theta - 60^\circ)}$$

$$\Rightarrow \frac{\tan(\theta + 60^\circ)}{\tan(\theta - 60^\circ)} = 3$$

$$\Rightarrow \frac{\tan(\theta + 60^\circ)}{\tan(\theta - 60^\circ)} = \frac{3}{1}$$

$$\Rightarrow \frac{\tan(\theta + 60^\circ) + \tan(\theta - 60^\circ)}{\tan(\theta + 60^\circ) - \tan(\theta - 60^\circ)} = \frac{3+1}{3-1}$$

$$\Rightarrow \frac{\sin(\theta + 60^\circ + \theta - 60^\circ)}{\sin(\theta + 60^\circ - \theta + 60^\circ)} = 2$$

$$\Rightarrow \frac{\sin(2\theta)}{\sin(120^\circ)} = 2$$

$$\Rightarrow \sin(2\theta) = 2 \sin(120^\circ)$$

$$\Rightarrow \sin(2\theta) = 2 \times \frac{\sqrt{3}}{2} = \sqrt{3}$$

It is not possible.

Hence, the equation has no solution.

18. Given equation is

$$\tan \theta + \tan 2\theta + \tan 3\theta = 0$$

$$\Rightarrow \tan \theta + \tan 2\theta + \tan(2\theta + \theta) = 0$$

$$\Rightarrow \tan \theta + \tan 2\theta + \frac{\tan(2\theta) + \tan(\theta)}{1 - \tan(2\theta)\tan(\theta)} = 0$$

$$\Rightarrow (\tan \theta + \tan 2\theta) \left(1 + \frac{1}{1 - \tan(2\theta)\tan(\theta)} \right) = 0$$

$$\Rightarrow (\tan \theta + \tan 2\theta) = 0, \left(1 + \frac{1}{1 - \tan(2\theta)\tan(\theta)} \right) = 0$$

when $(\tan \theta + \tan 2\theta) = 0$

$$\Rightarrow \tan \theta + \frac{2 \tan \theta}{1 - \tan^2 \theta} = 0$$

$$\Rightarrow \tan \theta \left(1 + \frac{2}{1 - \tan^2 \theta} \right) = 0$$

$$\Rightarrow \tan \theta = 0, \left(1 + \frac{2}{1 - \tan^2 \theta} \right) = 0$$

$$\Rightarrow \tan \theta = 0, \frac{2}{1 - \tan^2 \theta} = -1$$

$$\Rightarrow \tan \theta = 0, 1 - \tan^2 \theta = -2$$

$$\Rightarrow \tan \theta = 0, \tan^2 \theta = 3$$

$$\Rightarrow \theta = n\pi, \theta = n\pi \pm \frac{\pi}{3}, n \in I$$

$$\text{when } \left(1 + \frac{1}{1 - \tan(2\theta)\tan(\theta)} \right) = 0$$

$$\Rightarrow \frac{1}{1 - \tan \theta \tan 2\theta} = -1$$

$$\Rightarrow 1 - \tan \theta \tan 2\theta = -1$$

$$\Rightarrow \tan \theta \tan 2\theta = 2$$

$$\Rightarrow \tan \theta \left(\frac{2 \tan \theta}{1 - \tan^2 \theta} \right) = 2$$

$$\Rightarrow \tan^2 \theta = 1 - \tan^2 \theta$$

$$\Rightarrow \tan^2 \theta = \frac{1}{2} = \tan^2 \alpha, \alpha = \tan^{-1} \left(\frac{1}{\sqrt{2}} \right)$$

$$\Rightarrow \theta = n\pi \pm \alpha, n \in I$$

19. Given equation is

$$\Rightarrow \cos 2\theta \cos 4\theta = \frac{1}{2}$$

$$\Rightarrow 2 \cos(4\theta) \cos(2\theta) = 1$$

$$\Rightarrow \cos(6\theta) + \cos(2\theta) = 1$$

$$\Rightarrow \cos(6\theta) = 1 - \cos(2\theta)$$

20. Given equation is

$$\cot \theta - \tan \theta = \cos \theta - \sin \theta$$

$$\Rightarrow (\cos \theta - \sin \theta) \left(\frac{(\cos \theta + \sin \theta)}{\sin \theta \cos \theta} - 1 \right) = 0$$

$$\Rightarrow (\cos \theta - \sin \theta) = 0, (\cos \theta + \sin \theta) = \sin \theta \cos \theta$$

$$\Rightarrow \tan \theta = 1, (\cos \theta + \sin \theta) = \sin \theta \cos \theta$$

when $\tan \theta = 1$

$$\Rightarrow \theta = n\pi + \frac{\pi}{4}, n \in I$$

when $(\cos \theta + \sin \theta) = \sin \theta \cos \theta$

No real value of θ satisfies the given equation.

21. Given equation is

$$(1 - \tan \theta)(1 + \sin 2\theta) = 1 + \tan \theta$$

$$\Rightarrow (\cos \theta - \sin \theta)(\cos \theta + \sin \theta)^2 = (\cos \theta + \sin \theta)$$

$$\Rightarrow (\cos \theta + \sin \theta)(\cos 2\theta - 1) = 0$$

$$\Rightarrow \tan(\theta) = -1, \sin^2 \theta = 0$$

$$\Rightarrow \tan(\theta) = -1, \sin(\theta) = 0$$

$$\Rightarrow \theta = n\pi - \frac{\pi}{4}, \theta = n\pi, n \in I$$

22. Given equation is

$$2 \sin^2 \theta + \sin^2 2\theta = 2$$

$$\Rightarrow 2 \sin^2 \theta + 4 \sin^2 \theta \cos^2 \theta = 2$$

$$\Rightarrow \sin^2 \theta + 2 \sin^2 \theta \cos^2 \theta = 1$$

$$\Rightarrow 2 \sin^2 \theta \cos^2 \theta = 1 - \sin^2 \theta$$

$$\begin{aligned} \Rightarrow 2 \sin^2 \theta \cos^2 \theta &= \cos^2 \theta \\ \Rightarrow (2 \sin^2 \theta - 1) \cos^2 \theta &= 0 \\ \Rightarrow (2 \sin^2 \theta - 1) = 0, \cos^2 \theta &= 0 \end{aligned}$$

$$\Rightarrow \sin^2 \theta = \frac{1}{2}, \cos \theta = 0$$

$$\Rightarrow \theta = (2n+1)\frac{\pi}{2}, \theta = n\pi \pm \frac{\pi}{4}, n \in I$$

23. Given equation is

$$\begin{aligned} \sin(3\alpha) &= 4 \sin \theta \sin(\theta + \alpha) \sin(\theta + \alpha) \\ \Rightarrow \sin(3\alpha) &= 4 \sin \theta (\sin^2 \theta - \sin^2 \alpha) \\ \Rightarrow 3 \sin \alpha - 4 \sin^3 \alpha &= 4 \sin \alpha (\sin^2 \theta - \sin^2 \alpha) \end{aligned}$$

It is possible only when

$$\sin^2 \theta = \frac{3}{4}$$

$$\Rightarrow \sin^2 \theta = \left(\frac{\sqrt{3}}{2}\right)^2$$

$$\Rightarrow \theta = n\pi \pm \frac{\pi}{3}, n \in I$$

24. Given equation is

$$\begin{aligned} 4 \sin \theta \sin 2\theta \sin 4\theta &= \sin 3\theta \\ \Rightarrow 4 \sin \theta \sin(3\theta - \theta) \sin(3\theta + \theta) &= \sin 3\theta \\ \Rightarrow 4 \sin \theta [\sin^2(3\theta) - \sin^2(\theta)] &= \sin 3\theta \\ \Rightarrow 4 \sin \theta [\sin^2(3\theta) - \sin^2(\theta)] &= 3 \sin \theta - 4 \sin^3 \theta \\ \Rightarrow \sin \theta [4 \sin^2(3\theta) - 4 \sin^2(\theta) + 4 \sin^2 \theta - 3] &= 0 \\ \Rightarrow \sin \theta [4 \sin^2(3\theta) - 3] &= 0 \\ \Rightarrow \sin \theta = 0, [4 \sin^2(3\theta) - 3] &= 0 \end{aligned}$$

$$\Rightarrow \sin \theta = 0, \sin^2(3\theta) = \frac{3}{4}$$

$$\Rightarrow \theta = n\pi, \theta = n\pi \pm \frac{\pi}{3}, n \in I$$

25. We have $\sin(\theta) + \cos(\theta) = 1$

$$\Rightarrow \sqrt{2} \left(\frac{1}{\sqrt{2}} \sin(\theta) + \frac{1}{\sqrt{2}} \cos(\theta) \right) = 1$$

$$\Rightarrow \left(\frac{1}{\sqrt{2}} \sin(\theta) + \frac{1}{\sqrt{2}} \cos(\theta) \right) = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \sin \left(\theta + \frac{\pi}{4} \right) = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \sin \left(\theta + \frac{\pi}{4} \right) = \sin \left(\frac{\pi}{4} \right)$$

$$\Rightarrow \left(\theta + \frac{\pi}{4} \right) = \left(n\pi + (-1)^n \left(\frac{\pi}{4} \right) \right)$$

$$\Rightarrow \theta = \left(n\pi + (-1)^n \left(\frac{\pi}{4} \right) - \frac{\pi}{4} \right), n \in I$$

26. We have $\sqrt{3} \sin(\theta) + \cos(\theta) = 2$

$$\Rightarrow \frac{\sqrt{3}}{2} \sin \theta + \frac{1}{2} \cos \theta = 1$$

$$\Rightarrow \sin \left(\theta + \frac{\pi}{6} \right) = 1$$

$$\Rightarrow \sin \left(\theta + \frac{\pi}{6} \right) = \sin \left(\frac{\pi}{2} \right)$$

$$\Rightarrow \left(\theta + \frac{\pi}{6} \right) = n\pi + (-1)^n \left(\frac{\pi}{2} \right)$$

$$\Rightarrow \theta = n\pi + (-1)^n \left(\frac{\pi}{2} \right) - \frac{\pi}{6}$$

27. We have

$$\begin{aligned} \sin(2\theta) + \cos(2\theta) + \sin(\theta) + \cos(\theta) + 1 &= 0 \\ \Rightarrow (\sin(\theta) + \cos(\theta)) + (1 + \sin(2\theta)) + \cos(2\theta) &= 0 \\ \Rightarrow (\sin(\theta) + \cos(\theta)) + (\sin(\theta) + \cos(\theta))^2 &+ (\cos^2 \theta - \sin^2 \theta) = 0 \\ \Rightarrow (\sin(\theta) + \cos(\theta)) + (\sin(\theta) + \cos(\theta))^2 &+ (\cos \theta + \sin \theta)(\cos \theta - \sin \theta) = 0 \\ \Rightarrow (\sin(\theta) + \cos(\theta)) &= 0 \\ (1 + (\sin(\theta) + \cos(\theta)) + (\cos \theta - \sin \theta)) &= 0 \\ \Rightarrow (\sin(\theta) + \cos(\theta))(1 + 2 \cos \theta) &= 0 \\ \Rightarrow (\sin(\theta) + \cos(\theta)) = 0 \text{ and } (1 + 2 \cos \theta) &= 0 \end{aligned}$$

$$\Rightarrow \left(\sin \left(\frac{\pi}{4} + \theta \right) \right) = 0 \text{ and } \cos \theta = -\frac{1}{2}$$

$$\Rightarrow \left(\frac{\pi}{4} + \theta \right) = n\pi \text{ and } \theta = 2n\pi \pm \left(\frac{2\pi}{3} \right)$$

$$\Rightarrow \theta = n\pi - \frac{\pi}{4} \text{ and } \theta = 2n\pi \pm \left(\frac{2\pi}{3} \right), n \in I$$

28. We have $\sin^2 \theta + \sin \theta \cos \theta + \cos^2 \theta = 1$

$$\begin{aligned} \Rightarrow (\sin^3 \theta + \cos^3 \theta) + \sin \theta \cos \theta &= 1 \\ \Rightarrow (\sin \theta + \cos \theta)(1 - \sin \theta \cos \theta) + \sin \theta \cos \theta &= 1 \\ \Rightarrow (\sin \theta + \cos \theta)(1 - \sin \theta \cos \theta) &= (1 - \sin \theta \cos \theta) \\ \Rightarrow (\sin \theta + \cos \theta - 1)(1 - \sin \theta \cos \theta) &= 0 \\ \Rightarrow (\sin \theta + \cos \theta - 1) = 0 \text{ and } (1 - \sin \theta \cos \theta) &= 0 \end{aligned}$$

$$\Rightarrow (\sin \theta + \cos \theta) = 1 \text{ and } \sin(2\theta) = \frac{1}{2}$$

$$\Rightarrow \left(\sin \left(\theta + \frac{\pi}{4} \right) \right) = \sin \left(\frac{\pi}{2} \right)$$

$$\text{and } \sin(2\theta) = \sin \left(\frac{\pi}{6} \right)$$

$$\Rightarrow \theta = n\pi + (-1)^n \left(\frac{\pi}{2} \right) - \left(\frac{\pi}{4} \right)$$

$$\text{and } \theta = \frac{1}{2} \left(n\pi + (-1)^n \left(\frac{\pi}{6} \right) \right), \text{ where } n \in I$$

29. Given equation is

$$\sin \theta + \sqrt{3} \cos \theta = \sqrt{2}$$

$$\Rightarrow \frac{1}{2} \sin \theta + \frac{\sqrt{3}}{2} \cos \theta = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \sin\left(\theta + \frac{\pi}{3}\right) = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \left(\theta + \frac{\pi}{3}\right) = n\pi + (-1)^n \left(\frac{\pi}{4}\right)$$

$$\Rightarrow \theta = n\pi + (-1)^n \left(\frac{\pi}{4}\right) - \frac{\pi}{3}, n \in I$$

30. Do yourself.

31. Do yourself.

32. Do yourself.

33. Do yourself.

34. Do yourself.

35. Do yourself.

36. Do yourself.

37. Given equation is

$$\cos \theta + \sqrt{3} \sin \theta = 2 \cos 2\theta$$

$$\Rightarrow \frac{1}{2} \cos \theta + \frac{\sqrt{3}}{2} \sin \theta = \cos 2\theta$$

$$\Rightarrow \cos\left(\theta - \frac{\pi}{3}\right) = \cos 2\theta$$

$$\Rightarrow \left(\theta - \frac{\pi}{3}\right) = 2n\pi \pm 2\theta$$

Taking positive one, we get

$$\theta = -\left(2n\pi + \frac{\pi}{3}\right)$$

Taking negative one, we get,

$$\Rightarrow \theta = \frac{2n\pi}{3} + \frac{\pi}{9}, n \in I$$

38. Given equation is

$$\sqrt{3}(\cos \theta - \sqrt{3} \sin \theta) = 4 \sin 2\theta \cdot \cos 3\theta$$

$$\Rightarrow \sqrt{3} \cos \theta - 3 \sin \theta = 2(\sin 5\theta - \sin \theta)$$

$$\Rightarrow \sqrt{3} \cos \theta - \sin \theta = 2(\sin 5\theta)$$

$$\Rightarrow \frac{\sqrt{3}}{2} \cos \theta - \frac{1}{2} \sin \theta = (\sin 5\theta)$$

$$\Rightarrow \sin\left(\frac{\pi}{3} - \theta\right) = \sin 5\theta$$

$$\Rightarrow 5\theta = n\pi + (-1)^n \left(\frac{\pi}{3} - \theta\right)$$

when n is even

$$5\theta = 2k\pi + \left(\frac{\pi}{3} - \theta\right)$$

$$\Rightarrow 6\theta = 2k\pi + \frac{\pi}{3}$$

$$\Rightarrow \theta = \frac{k\pi}{3} + \frac{\pi}{18}, k \in I$$

when n is odd

$$5\theta = (2k+1)\pi - \left(\frac{\pi}{3} - \theta\right)$$

$$\Rightarrow 4\theta = (2k+1)\pi - \frac{\pi}{3}$$

$$\Rightarrow \theta = (2k+1)\frac{\pi}{4} - \frac{\pi}{12}, k \in I$$

$$39. \text{ We have } \sin(\theta) = -\frac{1}{2}$$

$$\Rightarrow \theta = -\frac{\pi}{6}$$

Hence, the principal value of θ is $\left(-\frac{\pi}{6}\right)$

$$40. \text{ We have } \sin(\theta) = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \theta = \frac{\pi}{4}, \frac{3\pi}{4}$$

Hence the principal value of θ is $\frac{\pi}{4}$

$$41. \text{ We have } \tan(\theta) = -\sqrt{3}$$

$$\Rightarrow \theta = -\frac{\pi}{3}$$

Hence, the principal value of θ is $-\frac{\pi}{3}$

$$42. \text{ Given, } \tan \theta = -1$$

$$\Rightarrow \theta = \frac{3\pi}{4}, -\frac{\pi}{4}$$

Hence, the principal value of θ is $-\frac{\pi}{4}$

$$43. \text{ Given, } \cos \theta = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

Hence, the principal value of θ is $\frac{\pi}{3}$

$$44. \text{ Given, } \cos \theta = -\frac{1}{2}$$

$$\Rightarrow \theta = \frac{2\pi}{3}, \frac{4\pi}{3}$$

Hence, the principal value of θ is $\frac{2\pi}{3}$

$$45. \text{ Given, } \tan \theta = -\sqrt{3}$$

$$\Rightarrow \theta = \frac{2\pi}{3}, -\frac{\pi}{3}$$

Hence, the principal value of θ is $-\frac{\pi}{3}$.

46. Given, $\sec \theta = \sqrt{2}$.

$$\Rightarrow \cos \theta = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \theta = \frac{\pi}{4}, \frac{7\pi}{4}$$

Hence, the principal value of θ is $\frac{\pi}{4}$.

47. Now $\sin(\theta) = \frac{1}{\sqrt{2}}$

$$\Rightarrow \theta = \frac{\pi}{4}, \frac{3\pi}{4}$$

and $\cos(\theta) = -\frac{1}{\sqrt{2}}$

$$\Rightarrow \theta = \frac{3\pi}{4}, \frac{5\pi}{4}$$

Thus, the common value of θ is $\frac{3\pi}{4}$.

Hence, the general values of θ is

$$\left(2n\pi + \frac{3\pi}{4}\right)$$

48. We have $\sin(\theta) = \frac{1}{\sqrt{2}}$

$$\Rightarrow \theta = \frac{\pi}{4}, \frac{3\pi}{4}$$

Also, $\tan \theta = -1$

$$\Rightarrow \theta = \frac{3\pi}{4}, \frac{7\pi}{4}$$

Thus, the common value of θ is $\frac{3\pi}{4}$.

Hence, the general values of θ is $\left(2n\pi + \frac{3\pi}{4}\right)$, where $n \in I$.

49. Given, $\cos \theta = \frac{1}{\sqrt{2}}$

$$\Rightarrow \theta = \frac{\pi}{4}, \frac{7\pi}{4}$$

and $\tan \theta = -1$

$$\Rightarrow \theta = \frac{3\pi}{4}, \frac{7\pi}{4}$$

Hence, the general solution is

$$\theta = 2n\pi + \frac{7\pi}{4}, n \in I$$

50. Given, $\sin \theta = \frac{1}{2}$

$$\Rightarrow \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

and $\tan \theta = \frac{1}{\sqrt{3}}$

$$\Rightarrow \theta = \frac{\pi}{6}, \frac{4\pi}{6}$$

Hence, the general solution is

$$\theta = 2n\pi + \frac{\pi}{6}, n \in I$$

51. We have, $(1 + \tan A)(1 + \tan B) = 2$

$$\Rightarrow 1 + \tan A + \tan B + \tan A \cdot \tan B = 2$$

$$\Rightarrow \tan A + \tan B = 1 - \tan A \cdot \tan B$$

$$\Rightarrow \left(\frac{\tan A + \tan B}{1 - \tan A \cdot \tan B}\right) = 1$$

$$\Rightarrow \tan(A + B) = 1$$

$$\Rightarrow \tan(A + B) = \tan\left(\frac{\pi}{4}\right)$$

$$\Rightarrow (A + B) = n\pi + \left(\frac{\pi}{4}\right), \text{ where } n \in I$$

52. Given, $\tan(A - B) = 1$

$$\Rightarrow (A - B) = \frac{\pi}{4}, \frac{5\pi}{4}$$

Also, $\sec(A + B) = \frac{2}{\sqrt{3}}$

$$\Rightarrow \cos(A + B) = \frac{\sqrt{3}}{2}$$

$$\Rightarrow (A + B) = \frac{\pi}{6}, \frac{11\pi}{6}$$

Here, we observe that $A - B$ is positive

So, $A > B$

$$\Rightarrow A + B > A - B$$

$$\begin{cases} A + B = \frac{11\pi}{6} \\ A - B = \frac{\pi}{4} \end{cases} \text{ or } \begin{cases} A + B = \frac{11\pi}{6} \\ A - B = \frac{5\pi}{4} \end{cases}$$

On solving, we get,

$$\begin{cases} A = \frac{25\pi}{24} \\ B = \frac{19\pi}{24} \end{cases} \text{ or } \begin{cases} A = \frac{19\pi}{24} \\ B = \frac{7\pi}{24} \end{cases}$$

General values of $\tan(A - B) = 1$

is $(A - B) = n\pi + \frac{\pi}{4}, n \in I \quad \dots(i)$

$$\text{General values of } \sec(A+B) = \frac{2}{\sqrt{3}}$$

$$\text{is } (A+B) = 2n\pi + \frac{\pi}{6}, n \in I \quad \dots(ii)$$

On solving (i) and (ii), we get

$$\begin{cases} A = (2n+m)\frac{\pi}{2} + \frac{\pi}{24} \\ B = (2n-m)\frac{\pi}{2} - \frac{5\pi}{24} \end{cases}$$

53. We have $\sin(\pi \cos \theta) = \cos(\pi \sin \theta)$

$$\Rightarrow \sin(\pi \cos \theta) = \sin\left(\frac{\pi}{2} - \pi \sin \theta\right)$$

$$\Rightarrow (\pi \cos \theta) = \left(\frac{\pi}{2} - \pi \sin \theta\right)$$

$$\Rightarrow \cos \theta = \left(\frac{1}{2} - \sin \theta\right)$$

$$\Rightarrow \cos \theta + \sin \theta = \frac{1}{2}$$

$$\Rightarrow \frac{1}{\sqrt{2}} \cos \theta + \frac{1}{\sqrt{2}} \sin \theta = \frac{1}{2\sqrt{2}}$$

$$\Rightarrow \cos\left(\theta + \frac{\pi}{4}\right) = \frac{1}{2\sqrt{2}}$$

Similarly, we can prove that,

$$\cos\left(\theta - \frac{\pi}{4}\right) = \frac{1}{2\sqrt{2}}$$

54. We have $\tan(\pi \cos \theta) = \cot(\pi \sin \theta)$

$$\Rightarrow \tan(\pi \cos \theta) = \tan\left(\frac{\pi}{2} - \pi \sin \theta\right)$$

$$\Rightarrow (\pi \cos \theta) = \left(\frac{\pi}{2} - \pi \sin \theta\right)$$

$$\Rightarrow \cos(\theta) + \sin(\theta) = \frac{1}{2}$$

$$\Rightarrow \frac{1}{\sqrt{2}} \cos(\theta) + \frac{1}{\sqrt{2}} \sin(\theta) = \frac{1}{2\sqrt{2}}$$

$$\Rightarrow \cos\left(\theta - \frac{\pi}{4}\right) = \frac{1}{2\sqrt{2}}$$

55. Given, $\sin A = \sin B$...(i)

and $\cos A = \cos B$...(ii)

Dividing (i) and (ii), we get,

$$\frac{\sin A}{\cos A} = \frac{\sin B}{\cos B}$$

$$\Rightarrow \tan A = \tan B$$

$$\Rightarrow A = n\pi + B, \text{ where } n \in I$$

56. Given equations are

$$3 \sin^2 A + 2 \sin^2 B = 1 \quad \dots(i)$$

$$\text{and } 3 \sin 2A - 2 \sin 2B = 0 \quad \dots(ii)$$

From (ii), we get,

$$3 \sin 2A = 2 \sin 2B$$

$$\Rightarrow \frac{\sin 2A}{2} = \frac{\sin 2B}{3}$$

$$\Rightarrow \frac{\sin 2B}{\sin 2A} = \frac{3}{2}$$

From (i), we get

$$\frac{3}{2}(2 \sin^2 A) + (2 \sin^2 B) = 1$$

$$\Rightarrow \frac{3}{2}(1 - \cos 2A) + (1 - \cos 2B) = 1$$

$$\Rightarrow \frac{3}{2} \cos 2A + \cos 2B = \frac{3}{2}$$

$$\Rightarrow \frac{\sin 2B}{\sin 2A} \cos 2A + \cos 2B = \frac{\sin 2B}{\sin 2A}$$

$$\Rightarrow \sin 2B \cos 2A + \sin 2A \cos 2B = \sin 2B$$

$$\Rightarrow \sin(2A + 2B) = \sin 2B$$

$$\Rightarrow \sin(2A + 2B) = \sin(\pi - 2B)$$

$$\Rightarrow (2A + 2) = (\pi - 2B)$$

$$\Rightarrow (2A + 4B) = \pi$$

$$\Rightarrow (A + 2B) = \frac{\pi}{2}$$

57. Given, $x + y = \frac{\pi}{4}$ and $\tan x + \tan y = 1$

$$\Rightarrow \tan(x + y) = \tan\left(\frac{\pi}{4}\right)$$

$$\Rightarrow \frac{\tan x + \tan y}{1 - \tan x \cdot \tan y} = 1$$

$$\Rightarrow 1 - \tan x \cdot \tan y = 1$$

$$\Rightarrow \tan x \cdot \tan y = 0$$

$$\Rightarrow \tan x = 0 \text{ \& \; } \tan y = 0$$

$$\Rightarrow x = n\pi = y$$

Thus, no values of x and y satisfy the given equations.

Therefore, the given equations have no solutions.

58. Given, $\sin x + \sin y = 1$...(i)

and $\cos 2x - \cos 2y = 1$...(ii)

From (ii), we get, $\cos 2x - \cos 2y = 1$

$$\Rightarrow 1 - 2 \sin^2 x - 1 + \sin^2 y = 1$$

$$\Rightarrow \sin^2 x - \sin^2 y = -\frac{1}{2}$$

$$\Rightarrow y = n\pi + (-1)^n \sin^{-1}\left(\frac{3}{4}\right),$$

where $n \in I$

$$\Rightarrow \sin x - \sin y = -\frac{1}{2}$$

Adding (i) and (iii), we get,

$$2 \sin x = \frac{1}{2}$$

$$\Rightarrow \sin x = \frac{1}{4}$$

$$\Rightarrow x = n\pi + (-1)^n \sin^{-1}\left(\frac{1}{4}\right), n \in I,$$

Subtracting (i) and (iii), we get

$$2 \sin y = \frac{3}{2}$$

$$\Rightarrow \sin y = \frac{3}{4}$$

$$\Rightarrow y = n\pi + (-1)^n \sin^{-1}\left(\frac{3}{4}\right), n \in I$$

59. Given, $\sin x = 2 \sin y$

$$\Rightarrow \sin x = 2 \sin\left(\frac{2\pi}{3} - x\right)$$

$$\Rightarrow \sin x = 2\left(\frac{\sqrt{3}}{2} \cos x + \frac{1}{2} \sin x\right)$$

$$\Rightarrow \sin x = \sqrt{3} \cos x + \sin x$$

$$\Rightarrow \sqrt{3} \cos x = 0$$

$$\Rightarrow \cos x = 0$$

$$\Rightarrow x = (2n+1)\frac{\pi}{2}$$

$$\text{when } x = (2n+1)\frac{\pi}{2}, \text{ then } y = n\pi - \frac{\pi}{6}$$

Hence, the solutions are

$$\begin{cases} x = (2n+1)\frac{\pi}{2} \\ y = n\pi - \frac{\pi}{6} \end{cases}, n \in I$$

60. Given, $x + y = \frac{2\pi}{3}$ and $\cos x + \cos y = \frac{3}{2}$

$$\text{Now } \cos x + \cos y = \frac{3}{2}$$

$$\Rightarrow \cos x + \cos\left(\frac{2\pi}{3} - x\right) = \frac{3}{2}$$

$$\Rightarrow \cos x - \frac{1}{2} \cos x + \frac{\sqrt{3}}{2} \sin x = \frac{3}{2}$$

$$\Rightarrow \frac{1}{2} \cos x + \frac{\sqrt{3}}{2} \sin x = \frac{3}{2}$$

$$\Rightarrow \cos x + \sqrt{3} \sin x = 3$$

...(iii)

It is not possible, since the maximum value of LHS is 2.

So, the given system of equations has no solutions.

61. Given equations are

$$r \sin \theta = 3 \quad \dots(i)$$

$$\text{and } r = 4(1 + \sin \theta) \quad \dots(ii)$$

Eliminating (i) and (ii), we get

$$4(1 + \sin \theta) \sin \theta = 3$$

$$\Rightarrow 4 \sin^2 \theta + 4 \sin \theta - 3 = 0$$

$$\Rightarrow 4 \sin^2 \theta + 6 \sin \theta - 2 \sin \theta - 3 = 0$$

$$\Rightarrow 2 \sin \theta (2 \sin \theta + 3) - 1(2 \sin \theta + 3) = 0$$

$$\Rightarrow (2 \sin \theta + 3)(2 \sin \theta - 1) = 0$$

$$\Rightarrow \sin \theta = -\frac{3}{2}, \frac{1}{2}$$

$$\Rightarrow \sin \theta = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

62. Given equations are

$$\sin x + \sin y = 1 \quad \dots(i)$$

$$\text{and } \cos 2x - \cos 2y = 1$$

$$\text{Now, } \cos 2x - \cos 2y = 1$$

$$\Rightarrow 1 - 2 \sin^2 x - 1 + 2 \sin^2 y = 1$$

$$\Rightarrow -2 \sin^2 x - 1 + 2 \sin^2 y = 0$$

$$\Rightarrow 2(\sin^2 x - \sin^2 y) = -1$$

$$\Rightarrow (\sin x + \sin y)(\sin x - \sin y) = -\frac{1}{2}$$

$$\Rightarrow (\sin x - \sin y) = -\frac{1}{2} \quad \dots(ii)$$

On solving, we get

$$\sin x = 0, \sin y = 1$$

$$\Rightarrow x = n\pi, y = (4n+1)\frac{\pi}{2}, n \in I$$

Hence, the solutions are

$$\begin{cases} x = n\pi \\ y = (4n+1)\frac{\pi}{2} \end{cases}, n \in I$$

63. Given curves are $y = \cos x$ and $y = \sin 2x$

$$\text{Thus, } \sin 2x = \cos x$$

$$\Rightarrow 2 \sin x \cos x = \cos x$$

$$\Rightarrow (2 \sin x - 1) \cos x = 0$$

$$\Rightarrow (2 \sin x - 1) = 0, \cos x = 0$$

$$\Rightarrow \sin x = \frac{1}{2}, \cos x = 0$$

$$\Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}, \frac{3\pi}{2}$$

$$\text{then } y = \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}, 0$$

Hence, the solutions are

$$\left(\frac{\pi}{6}, \frac{\sqrt{3}}{2}\right), \left(\frac{5\pi}{6}, -\frac{\sqrt{3}}{2}\right), \left(\frac{\pi}{2}, 0\right), \left(\frac{3\pi}{2}, 0\right)$$

64. Given equation is

$$\begin{aligned} \Rightarrow \cos x + \cos y + \cos(x+y) &= -\frac{3}{2} \\ \Rightarrow 2(\cos x + \cos y) + 2\cos(x+y) &= -1 \\ \Rightarrow 4\cos\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right) + 4\cos^2\left(\frac{x+y}{2}\right) &= -1 \\ \Rightarrow 4\cos^2\left(\frac{x+y}{2}\right) + 4\cos\left(\frac{x-y}{2}\right)\cos\left(\frac{x+y}{2}\right) + 1 &= 0 \end{aligned}$$

For real x and y ,

$$16\cos^2\left(\frac{x-y}{2}\right) - 16 \geq 0$$

$$\Rightarrow \cos^2\left(\frac{x-y}{2}\right) \geq 1$$

$$\Rightarrow \cos^2\left(\frac{x-y}{2}\right) = 1$$

$$\Rightarrow \left(\frac{x-y}{2}\right) = 0$$

$$\Rightarrow x = y$$

The given equation

$$4\cos^2\left(\frac{x+y}{2}\right) + 4\cos\left(\frac{x-y}{2}\right)\cos\left(\frac{x+y}{2}\right) + 1 = 0$$

reduces to $4\cos^2(x) + \cos(x) + 1 = 0$

$$\Rightarrow (2\cos(x) + 1)^2 = 0$$

$$\Rightarrow \cos(x) = -\frac{1}{2}$$

$$\Rightarrow x = \frac{2\pi}{3} = y$$

65. Given equation is

$$8\cos\theta\cos\phi\cos(\theta+\phi) + 1 = 0$$

$$\Rightarrow 2\cos\theta\cos\phi\cos(\theta+\phi) = -\frac{1}{4}$$

$$\Rightarrow 4[\cos(\theta+\phi) + \cos(\theta+\phi)]\cos(\theta+\phi) + 1 = 0$$

$$\Rightarrow 4\cos^2(\theta+\phi) + \cos(\theta+\phi)\cos(\theta+\phi) + 1 = 0$$

For all real $0 < \theta, \phi < \pi$,

$$16\cos^2(\theta-\phi) - 16 > 0$$

$$\Rightarrow \cos^2(\theta-\phi) \geq 1$$

$$\Rightarrow \cos^2(\theta-\phi) = 1$$

$$\Rightarrow \theta - \phi = 0$$

$$\Rightarrow \theta = \phi$$

when $\theta = \phi$, then the equation

$$4\cos^2(\theta+\phi) + 4\cos(\theta-\phi)\cos(\theta+\phi) + 1 = 0$$

reduces to

$$\Rightarrow 4\cos^2(2\theta) + 4\cos(2\theta) + 1 = 0$$

$$\Rightarrow (2\cos(2\theta) + 1)^2 = 0$$

$$\Rightarrow (2\cos(2\theta) + 1) = 0$$

$$\Rightarrow \cos(2\theta) = -\frac{1}{2}$$

$$\Rightarrow (2\theta) = \frac{2\pi}{3}$$

$$\Rightarrow \theta = \frac{\pi}{3} = \phi$$

$$66. \text{ We have } \frac{\tan 3x - \tan 2x}{1 + \tan 3x \cdot \tan 2x} = 1$$

$$\Rightarrow \tan(3x - 2x) = 1$$

$$\Rightarrow \tan x = 1$$

$$\Rightarrow x = n\pi + \frac{\pi}{4}, \text{ where } n \in I$$

But the values of x do not satisfy the given equation

Hence, the set of values of x is ϕ .

67. Given equation is $\tan x + \sec x = 2\cos x$

$$\Rightarrow (1 + \sin x) = 2\cos^2 x$$

$$\Rightarrow (1 + \sin x) = 2(1 - \sin^2 x)$$

$$\Rightarrow (1 + \sin x) = 2(1 + \sin x) \cdot (1 - \sin x)$$

$$\Rightarrow (1 + \sin x)(1 - 2 + 2\sin x) = 0$$

$$\Rightarrow (1 + \sin x)(2\sin x - 1) = 0$$

$$\Rightarrow (1 + \sin x) = 0 \text{ and } (2\sin x - 1) = 0$$

$$\Rightarrow \sin x = -1 \text{ and } \sin x = \frac{1}{2}$$

$$\Rightarrow x = \frac{\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$$

But $x = \frac{\pi}{2}$ does not satisfy the given equation.

Thus, the values of x are $\frac{\pi}{6}$ and $\frac{5\pi}{6}$.

Hence, the number of solutions is 2.

68. Given equation is $2\sin^2 x + 6\sin x - \sin x - 3 = 0$

$$\Rightarrow 2\sin^2 x + 6\sin x - \sin x - 3 = 0$$

$$\Rightarrow 2\sin x(\sin x + 3) - 1(\sin x + 3) = 0$$

$$\Rightarrow (\sin x + 3)(2\sin x - 1) = 0$$

$$\Rightarrow \sin x = -3, \frac{1}{2}$$

$$\Rightarrow \sin x = \frac{1}{2}$$

$$\Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}$$

Hence, the number of values of x is 4.

69. Given

$$\tan(x + 20^\circ) = \tan(x - 10^\circ) \tan x \cdot \tan(x + 10^\circ)$$

$$\Rightarrow \frac{\tan(x + 20^\circ)}{\tan x} = \tan(x - 10^\circ) \cdot \tan(x + 10^\circ)$$

$$\begin{aligned}
 &\Rightarrow \frac{\sin(x+20^\circ)\cos x}{\cos(x+20^\circ)\sin x} = \frac{\sin(x-10^\circ)\sin(x+10^\circ)}{\cos(x-10^\circ)\cos(x+10^\circ)} \\
 &\Rightarrow \frac{\sin(x+20^\circ)\cos x + \cos(x+20^\circ)\sin x}{\sin(x+20^\circ)\cos x - \cos(x+20^\circ)\sin x} \\
 &\quad = \frac{\sin(x-10^\circ)\sin(x+10^\circ) + \cos(x-10^\circ)\cos(x+10^\circ)}{\sin(x-10^\circ)\sin(x+10^\circ) - \cos(x-10^\circ)\cos(x+10^\circ)} \\
 &\Rightarrow \frac{\sin(x+20^\circ+x)}{\sin(x+20^\circ-x)} = \frac{\cos(x+10^\circ-x+10^\circ)}{\cos(x+10^\circ+x-10^\circ)} \\
 &\Rightarrow \frac{\sin(2x+20^\circ)}{\sin(20^\circ)} = \frac{\cos(20^\circ)}{\cos(2x)} \\
 &\Rightarrow \sin(2x+20^\circ)\cos(2x) = -\sin(20^\circ)\cos(20^\circ) \\
 &\Rightarrow 2\sin(2x+20^\circ)\cos(2x) = -2\sin(20^\circ)\cos(20^\circ) \\
 &\Rightarrow \sin(4x+20^\circ) + \sin(20^\circ) = \sin(40^\circ) \\
 &\Rightarrow \sin(4x+20^\circ) = \sin(40^\circ) - \sin(20^\circ) \\
 &\Rightarrow \sin(4x+20^\circ) = -2\sin(30^\circ)\cos(10^\circ) \\
 &\Rightarrow \sin(4x+20^\circ) = -\cos(10^\circ) \\
 &\Rightarrow \sin(4x+20^\circ) = -\sin(80^\circ) \\
 &\Rightarrow \sin(4x+20^\circ) = \sin(-80^\circ) \\
 &\Rightarrow \sin(4x+20^\circ) = \sin(\pi - (-80^\circ)) \\
 &\Rightarrow (4x+20^\circ) = (\pi - (-80^\circ)) \\
 &\Rightarrow (4x+20^\circ) = 260^\circ \\
 &\Rightarrow 4x = 260^\circ - 20^\circ = 240^\circ \\
 &\Rightarrow x = 60^\circ
 \end{aligned}$$

Hence, the smallest positive value of x is 60°

70. Given, $\sin^2 x + \cos^2 y = 2 \sec^2 z$

Here, LHS ≤ 2 and RHS ≥ 2

It is possible only when

$$\begin{aligned}
 &\sin^2 x = 1, \cos^2 y = 1, \sec^2 z = 1 \\
 &\Rightarrow \cos^2 x = 0, \sin^2 y = 0, \cos^2 z = 1 \\
 &\Rightarrow \cos^2 x = 0, \sin^2 y = 0, \sin^2 z = 1 \\
 &\Rightarrow \cos x = 0, \sin y = 0, \sin z = 0 \\
 &\Rightarrow x = (2n+1)\frac{\pi}{2}, y = m\pi, z = k\pi
 \end{aligned}$$

where, $n, m, k \in I$.

71. The given equation can be expressed as

$$\begin{aligned}
 &5(2\cos^2 x - 1) + (1 - \cos x) + 1 = 0 \\
 &\Rightarrow 10\cos^2 x + \cos x - 3 = 0 \\
 &\Rightarrow (5\cos x + 3)(2\cos x - 1) = 0 \\
 &\Rightarrow (5\cos x + 3) = 0, (2\cos x - 1) = 0 \\
 &\Rightarrow \cos x = -\frac{3}{5} = \cos \alpha, \\
 &\Rightarrow \cos x = \frac{1}{2} = \cos\left(\frac{\pi}{3}\right) \\
 &\Rightarrow x = 2n\pi \pm \alpha = 2n\pi \pm \cos^{-1}\left(\frac{\pi}{3}\right), \\
 &\Rightarrow x = 2n\pi \pm \frac{\pi}{3}, n \in Z
 \end{aligned}$$

72. Given equation is

$$\begin{aligned}
 &4\sin^4 x + \cos^4 x = 1 \\
 &\Rightarrow 4\sin^4 x = 1 - \cos^4 x \\
 &\Rightarrow 4\sin^4 x = (1 + \cos^2 x)\sin^2 x \\
 &\Rightarrow \sin^2 x (4\sin^2 x - \cos^2 x - 1) = 0 \\
 &\Rightarrow \sin^2 x = 0, (5\sin^2 x - 2) = 0 \\
 &\Rightarrow \sin x = 0, \sin^2 x = \frac{2}{5} \\
 &\Rightarrow x = n\pi, x = n\pi \pm \alpha, \alpha = \sin^{-1}\left(\sqrt{\frac{2}{5}}\right)
 \end{aligned}$$

73. Given equation is

$$\begin{aligned}
 &4\cos^2 x \sin x - 2\sin^2 x = 2\sin x \\
 &\Rightarrow 4(1 - \sin^2 x)\sin x - 2\sin^2 x = 2\sin x \\
 &\Rightarrow 2(1 - \sin^2 x)\sin x - \sin^2 x = \sin x \\
 &\Rightarrow 2\sin x - 2\sin^3 x - \sin^2 x - \sin x = 0 \\
 &\Rightarrow \sin x - 2\sin^3 x - \sin^2 x = 0 \\
 &\Rightarrow 2\sin^3 x + \sin^2 x - \sin x = 0 \\
 &\Rightarrow \sin x (2\sin^2 x + \sin x - 1) = 0 \\
 &\Rightarrow \sin x = 0, (2\sin^2 x + \sin x - 1) = 0 \\
 &\Rightarrow \sin x = 0, \sin x = \frac{-1 \pm 3}{2} \\
 &\Rightarrow \sin x = 0, \sin x = 1, \sin x = -2 \\
 &\Rightarrow \sin x = 0, \sin x = 1 \\
 &\Rightarrow x = n\pi, x = (4n+1)\frac{\pi}{2}, n \in I
 \end{aligned}$$

74. Given equation is

$$\begin{aligned}
 &\sin 3x + \cos 2x = 1 \\
 &\Rightarrow \sin 3x = 1 - \cos 2x \\
 &\Rightarrow \sin x (3 - 4\sin^2 x) = 2\sin^2 x \\
 &\Rightarrow \sin x (3 - 4\sin^2 x - 2\sin x) = 0 \\
 &\Rightarrow \sin x = 0, (4\sin^2 x + 2\sin x - 3) = 0 \\
 &\Rightarrow \sin x = 0, \sin x = \frac{-2 \pm \sqrt{4+48}}{8} \\
 &\Rightarrow \sin x = 0, \sin x = \frac{-1 \pm \sqrt{13}}{4} \\
 &\Rightarrow \sin x = 0, \sin x = \frac{\sqrt{13}-1}{4} \\
 &\Rightarrow x = n\pi, x = n\pi + (-1)^n \alpha, \alpha = \sin^{-1}\left(\frac{\sqrt{13}-1}{4}\right)
 \end{aligned}$$

75. Given equation is

$$\begin{aligned}
 &2\cos 2x + \sqrt{2\sin x} = 2 \\
 &\Rightarrow \sqrt{2}\sqrt{\sin x} = 2(1 - \cos 2x) \\
 &\Rightarrow \sqrt{2}\sqrt{\sin x} = 4\sin^2 x \\
 &\Rightarrow \sqrt{\sin x} = 2\sqrt{2}\sin^2 x \\
 &\Rightarrow \sqrt{\sin x} (1 - 2\sqrt{2}\sin^{3/2} x) = 0
 \end{aligned}$$

$$\Rightarrow \sqrt{\sin x} = 0, \sin^{3/2} x = \frac{1}{2\sqrt{2}}$$

$$\Rightarrow \sin x = 0, \sin x = \frac{1}{\sqrt{2}}$$

$$\Rightarrow x = n\pi, n = n\pi + (-1)^n \left(\frac{\pi}{4}\right), n \in I$$

76. Given equation is

$$1 + \sin^3 x + \cos^3 x = \frac{3}{2} \sin 2x$$

$$\Rightarrow 1 + (\sin^3 x + \cos^3 x) = 3 \sin x \cos x$$

$$\Rightarrow 1 + (\sin^3 x + \cos^3 x) - 3 \sin x \cos x \cdot 1 = 0$$

$$\Rightarrow (\sin x + \cos x + 1)(2 - \sin x \cos x - \sin x - \cos x) = 0$$

$$\Rightarrow (\sin x + \cos x + 1) = 0$$

$$\Rightarrow \sin x + \cos x = -1$$

$$\Rightarrow \frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x = -\frac{1}{\sqrt{2}}$$

$$\Rightarrow \sin \left(x + \frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$$

$$\Rightarrow \left(x + \frac{\pi}{4}\right) = n\pi + (-1)^n \left(-\frac{\pi}{4}\right)$$

$$\Rightarrow x = n\pi + (-1)^n \left(-\frac{\pi}{4}\right) - \frac{\pi}{4}, n \in I$$

77. Given equation is

$$\sin^6 x + \cos^6 x = \frac{7}{16}$$

$$\Rightarrow 1 - 3 \sin^2 x \cos^2 x = \frac{7}{16}$$

$$\Rightarrow 3 \sin^2 x \cos^2 x = 1 - \frac{7}{16} = \frac{9}{16}$$

$$\Rightarrow \sin^2 x \cos^2 x = \frac{3}{16}$$

$$\Rightarrow 4 \sin^2 x \cos^2 x = \frac{3}{4}$$

$$\Rightarrow \sin^2(2x) = \frac{3}{4}$$

$$\Rightarrow (2x) = n\pi \pm \frac{\pi}{3}$$

$$\Rightarrow x = \frac{n\pi}{3} \pm \frac{\pi}{6}, n \in I$$

78. Given equation is

$$\sin^8 x + \cos^8 x = \frac{17}{16} \cos^2 2x$$

$$\Rightarrow (\sin^4 x + \cos^4 x)^2 - 2 \sin^4 x \cos^4 x = \frac{17}{16} \cos^2 2x$$

$$\Rightarrow (1 - 2 \sin^2 x \cos^2 x)^2 - 2 \sin^4 x \cos^4 x = \frac{17}{16} \cos^2 2x$$

$$\Rightarrow (1 - 4 \sin^2 x \cos^2 x) + 2 \sin^4 x \cos^4 x = \frac{17}{16} \cos^2 2x$$

$$\Rightarrow 16(1 - 4 \sin^2 x \cos^2 x + 2 \sin^4 x \cos^4 x)$$

$$\Rightarrow 17(\cos^4 x + \sin^4 x - 2 \sin^2 x \cos^2 x)$$

$$\Rightarrow 17(1 - 4 \sin^2 x \cos^2 x)$$

$$\Rightarrow 32 \sin^4 x \cos^4 x + 4 \sin^2 x \cos^2 x - 1 = 0$$

$$\Rightarrow 2 \sin^4(2x) + \sin^2 2x - 1 = 0$$

$$\Rightarrow \sin^2(2x) = \frac{-1 \pm \sqrt{5}}{4}$$

$$\Rightarrow \sin^2(2x) = \frac{\sqrt{5} - 1}{4}$$

$$\Rightarrow 2x = n\pi \pm \alpha, \alpha = \sin^{-1} \left(\sqrt{\frac{\sqrt{5} - 1}{4}} \right)$$

$$\Rightarrow x = \frac{n\pi}{2} \pm \frac{\alpha}{2}, \alpha = \sin^{-1} \left(\sqrt{\frac{\sqrt{5} - 1}{4}} \right)$$

79. Given equation is

$$2 \sin^2 x + 2 = \cos^2 3x$$

$$\Rightarrow 2 \sin^3 x + 2 = 1 - \sin^2 3x$$

$$\Rightarrow 2 \sin^3 x + \sin^2 3x + 1 = 0$$

$$\Rightarrow 2 \sin^3 x + (3 \sin x - 4 \sin^3 x)^2 + 1 = 0$$

$$\Rightarrow 2 \sin^3 x + 9 \sin^2 x - 24 \sin^4 x + 16 \sin^6 x + 1 = 0$$

$$\Rightarrow 16 \sin^6 x - 24 \sin^4 x + 2 \sin^3 x + 9 \sin^2 x + 1 = 0$$

$$\Rightarrow \sin x = -1$$

$$x = (4n - 1) \frac{\pi}{2}, n \in I$$

80. Given equation is

$$\cos 4x = \cos^2 3x$$

$$\Rightarrow 2 \cos^2 2x - 1 = \cos^2 3x$$

$$\Rightarrow 2 \cos^2 2x = 1 + \cos^2 3x$$

It is possible only when

$$\cos^2 2x = 1, \cos^2 3x = 1$$

It is true for $x = 0$

Hence, the solution is $x = n\pi, n \in I$

81. Given equation is

$$\cos 2x = 6 \tan^2 x - 2 \cos^2 x$$

$$\Rightarrow 2 \cos^2 x - 1 = 6 \left(\frac{\sin^2 x}{\cos^2 x} \right) - 2 \cos^2 x$$

$$\Rightarrow 2 \cos^4 x - \cos^2 x = 6 - 6 \cos^2 x - 2 \cos^4 x$$

$$\Rightarrow 4 \cos^4 x + 5 \cos^2 x - 6 = 0$$

$$\Rightarrow 4 \cos^4 x + 8 \cos^2 x - 3 \cos^2 x - 6 = 0$$

$$\Rightarrow 4 \cos^2 x (\cos^2 x + 2) - 3(\cos^2 x + 2) = 0$$

$$\Rightarrow (4 \cos^2 x - 3)(\cos^2 x + 2) = 0$$

$$\Rightarrow (4 \cos^2 x - 3) = 0$$

$$\Rightarrow \cos^2 x = \frac{3}{4}$$

$$\Rightarrow x = n\pi \pm \frac{\pi}{6}, n \in I$$

82. The given equation can be written as

$$(2 \sin x - \cos x)(1 + \cos x) = (1 - \cos x)(1 + \cos x)$$

$$\Rightarrow (1 + \cos x)(2 \sin x - \cos x - 1 + \cos x) = 0$$

$$\Rightarrow (1 + \cos x)(2 \sin x - 1) = 0$$

$$\Rightarrow \cos x = -1, \sin x = 1/2$$

$$\Rightarrow \cos x = -1 = \cos \pi, \sin x = \frac{1}{2} = \sin\left(\frac{\pi}{6}\right)$$

$$\Rightarrow x = 2n\pi \pm \pi, x = n\pi + (-1)^n \frac{\pi}{6}, n \in \mathbb{Z}$$

83. Given equation is

$$2 \sin^2 x + \sin x - 1 = 0$$

$$\Rightarrow \sin x = \frac{-1 \pm 3}{4} = \frac{1}{2}, -1$$

$$\Rightarrow \sin x = \frac{1}{2}, \sin x = -1$$

$$\Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$$

84. Given equation is

$$5 \sin^2 x + 7 \sin x - 6 = 0$$

$$\Rightarrow 5 \sin^2 x + 10 \sin x - 3 \sin x - 6 = 0$$

$$\Rightarrow 5 \sin x (\sin x + 2) - 3(\sin x + 2) = 0$$

$$\Rightarrow (5 \sin x - 3)(\sin x + 2) = 0$$

$$\Rightarrow (5 \sin x - 3) = 0, (\sin x + 2) = 0$$

$$\Rightarrow (5 \sin x - 3) = 0$$

$$\Rightarrow \sin x = \frac{3}{5}$$

$$\Rightarrow x = n\pi + (-1)^n \alpha, \alpha = \sin^{-1}\left(\frac{3}{5}\right)$$

Hence, the solution is

$$x = \sin^{-1}\left(\frac{3}{5}\right), \pi - \sin^{-1}\left(\frac{3}{5}\right)$$

85. Given equation is

$$\sin^2 x - \cos x = \frac{1}{4}$$

$$\Rightarrow 4 \sin^2 x - 4 \cos x - 1 = 0$$

$$\Rightarrow 4 - 4 \cos^2 x - 4 \cos x - 1 = 0$$

$$\Rightarrow 3 - 4 \cos^2 x - 4 \cos x = 0$$

$$\Rightarrow 4 \cos^2 x + 4 \cos x - 3 = 0$$

$$\Rightarrow \cos x = \frac{-4 \pm 8}{8} = \frac{1}{2}, -\frac{3}{2}$$

$$\Rightarrow \cos x = \frac{1}{2}$$

$$\Rightarrow x = \frac{\pi}{3}, \frac{5\pi}{3}$$

86. Given equation is

$$\tan^2 x - 2 \tan x - 3 = 0$$

$$\Rightarrow (\tan x - 3)(\tan x + 1) = 0$$

$$\Rightarrow (\tan x - 3) = 0, (\tan x + 1) = 0$$

$$\Rightarrow (\tan x - 3) = 0, (\tan x + 1) = 0$$

$$\Rightarrow \tan x = -1, \tan x = 3$$

$$\Rightarrow x = n\pi - \frac{\pi}{4}, x = n\pi + \alpha, \alpha = \tan^{-1}(3)$$

87. Given equation is

$$2 \cos^2 x - \sqrt{3} \sin x + 1 = 0$$

$$\Rightarrow 2 - 2 \sin^2 x - \sqrt{3} \sin x + 1 = 0$$

$$\Rightarrow 3 - 2 \sin^2 x - \sqrt{3} \sin x = 0$$

$$\Rightarrow 2 \sin^2 x + \sqrt{3} \sin x - 3 = 0$$

$$\Rightarrow \sin x = \frac{-\sqrt{3} \pm \sqrt{27}}{4} = \frac{-\sqrt{3} \pm 3\sqrt{3}}{4}$$

$$\Rightarrow \sin x = \frac{-4\sqrt{3}}{4}, \frac{2\sqrt{3}}{4}$$

$$\Rightarrow \sin x = -\sqrt{3}, \frac{\sqrt{3}}{2}$$

$$\Rightarrow \sin x = \frac{\sqrt{3}}{2}$$

$$\Rightarrow x = n\pi + (-1)^n \left(\frac{\pi}{3}\right), n \in \mathbb{I}$$

88. The given equation can be written as

$$(\sin x + \sin 5x) + \sin 3x = 0$$

$$\Rightarrow 2 \sin 3x \cdot \cos 2x + \sin 3x = 0$$

$$\Rightarrow \sin 3x(2 \cos 2x + 1) = 0$$

$$\Rightarrow \sin 3x = 0, \cos 2x = -1/2$$

$$\Rightarrow \sin 3x = 0, \cos 2x = \cos\left(\frac{2\pi}{3}\right)$$

$$\Rightarrow 3x = n\pi, 2x = 2n\pi \pm \frac{2\pi}{3}, n \in \mathbb{Z}$$

$$\Rightarrow x = \frac{n\pi}{3}, x = n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$$

$$\Rightarrow x = 0, \frac{\pi}{3}$$

89. Given equation is

$$\cos x - \cos 2x = \sin 3x$$

$$\Rightarrow 2 \sin\left(\frac{3x}{2}\right) \sin\left(\frac{x}{2}\right) = 2 \sin\left(\frac{3x}{2}\right) \cos\left(\frac{3x}{2}\right)$$

$$\Rightarrow 2 \sin\left(\frac{3x}{2}\right) \left(\sin\left(\frac{x}{2}\right) - \cos\left(\frac{3x}{2}\right)\right) = 0$$

$$\Rightarrow \sin\left(\frac{3x}{2}\right) = 0, \left(\sin\left(\frac{x}{2}\right) - \cos\left(\frac{3x}{2}\right)\right) = 0$$

$$\text{when } \sin\left(\frac{3x}{2}\right) = 0$$

$$\text{Then } \frac{3x}{2} = n\pi$$

$$\Rightarrow x = \frac{2n\pi}{3}$$

$$\text{when } \left(\sin\left(\frac{x}{2}\right) - \cos\left(\frac{3x}{2}\right) \right) = 0$$

$$\Rightarrow \left(\sin\left(\frac{x}{2}\right) = \cos\left(\frac{3x}{2}\right) \right)$$

$$\Rightarrow \sin\left(\frac{x}{2}\right) = \cos\left(\frac{3x}{2}\right)$$

$$\Rightarrow \cos\left(\frac{3x}{2}\right) = \cos\left(\frac{\pi}{2} - \frac{x}{2}\right)$$

$$\Rightarrow \left(\frac{3x}{2} \right) = 2n\pi \pm \left(\frac{\pi}{2} - \frac{x}{2} \right)$$

Taking positive sign, we get,

$$2x = 2n\pi + \frac{\pi}{2}$$

$$\Rightarrow x = n\pi + \frac{\pi}{4}$$

Taking negative sign, we get,

$$x = 2n\pi - \frac{\pi}{2}, n \in I$$

90. Given equation is

$$\sin 7x + \sin 4x + \sin x = 0$$

$$\Rightarrow (\sin 7x + \sin x) + \sin 4x = 0$$

$$\Rightarrow 2 \sin 4x \cos 3x = \sin 4x = 0$$

$$\Rightarrow \sin 4x(2 \cos 3x + 1) = 0$$

$$\Rightarrow \sin 4x = 0, (2 \cos 3x + 1) = 0$$

$$\Rightarrow \sin 4x = 0, \cos 3x = -\frac{1}{2}$$

$$\Rightarrow 4x = n\pi, 3x = 2n\pi \pm \frac{2\pi}{3}$$

$$\Rightarrow x = \frac{n\pi}{4}, x = \frac{2n\pi}{3} \pm \frac{2\pi}{9}, n \in I$$

Hence, the solutions are

$$x = 0, \frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}$$

91. Given equation is

$$\cos 3x + \cos 2x = \sin\left(\frac{3x}{2}\right) + \sin\left(\frac{x}{2}\right)$$

$$\Rightarrow 2 \cos\left(\frac{5x}{2}\right) \cos\left(\frac{x}{2}\right) = 2 \sin x \cos\left(\frac{x}{2}\right)$$

$$\Rightarrow 2 \left(\cos\left(\frac{5x}{2}\right) - \sin x \right) \cos\left(\frac{x}{2}\right) = 0$$

$$\Rightarrow 2 \left(\cos\left(\frac{5x}{2}\right) - \sin x \right) = 0, \cos\left(\frac{x}{2}\right) = 0$$

$$\Rightarrow \cos\left(\frac{5x}{2}\right) = \sin x, \cos\left(\frac{x}{2}\right) = 0$$

$$\Rightarrow \cos\left(\frac{5x}{2}\right) = \cos\left(\frac{\pi}{2} - x\right), \cos\left(\frac{x}{2}\right) = 0$$

$$\Rightarrow \left(\frac{5x}{2} \right) = 2n\pi \pm \left(\frac{\pi}{2} - x \right), \left(\frac{x}{2} \right) = (2n+1)\frac{\pi}{2}$$

$$\Rightarrow x = \frac{4n\pi}{5} \pm (\pi - 2x), x = (2n+1)\pi$$

$$\Rightarrow x = \frac{4n\pi}{5} \pm (\pi - 2x), x = (2n+1)\pi$$

$$\Rightarrow x = \frac{\pi}{3}, \frac{\pi}{5}, \frac{3\pi}{5}, \frac{13\pi}{15}, \frac{17\pi}{15}, \frac{7\pi}{5}, \frac{5\pi}{3}, \frac{29\pi}{15}$$

92. Do yourself.

93. Given equation is

$$\cos 2x + \cos 4x = 2 \cos x$$

$$\Rightarrow 2 \cos 3x \cos x = 2 \cos x$$

$$\Rightarrow (2 \cos 3x - 1) \cos x = 0$$

$$\Rightarrow (2 \cos 3x - 1) = 0, \cos x = 0$$

$$\Rightarrow \cos 3x = \frac{1}{2}, \cos x = 0$$

$$\Rightarrow x = \frac{2n\pi}{3} \pm \frac{2\pi}{9}, x = (2n+1)\frac{\pi}{2}$$

94. Given equation is

$$\sin 2x + \cos 2x + \sin x + \cos x + 1 = 0$$

$$\Rightarrow (1 + \sin 2x) + (\sin x + \cos x) + \cos 2x = 0$$

$$\Rightarrow (\sin x + \cos x)^2 + (\sin x + \cos x) + (\cos^2 x - \sin^2 x) = 0$$

$$\Rightarrow (\sin x + \cos x)(2 \cos x + 1) = 0$$

$$\Rightarrow (\sin x + \cos x) = 0, (2 \cos x + 1) = 0$$

$$\Rightarrow \tan x = -1, \cos x = -\frac{1}{2}$$

$$\Rightarrow \tan x = -1, \cos x = -\frac{1}{2}$$

$$\Rightarrow x = n\pi - \frac{\pi}{4}, x = 2n\pi \pm \frac{2\pi}{3}, n \in I$$

95. Do yourself.

96. Given equation is

$$\tan 3x + \tan x = 2 \tan 2x$$

$$\Rightarrow \frac{\sin 4x}{\cos 3x \cos x} = \frac{2 \sin 2x}{\cos 2x}$$

$$\Rightarrow \frac{2 \sin 2x \cos 2x}{\cos 3x \cos x} = \frac{2 \sin 2x}{\cos 2x}$$

$$\Rightarrow 2 \sin 2x \left(\frac{\cos 2x}{\cos 3x \cos x} - \frac{1}{\cos 2x} \right) = 0$$

$$\Rightarrow 2 \sin 2x = 0, \left(\frac{\cos 2x}{\cos 3x \cos x} = \frac{1}{\cos 2x} \right)$$

$$\Rightarrow \sin 2x = 0, 2 \cos^2 2x = \cos 4x + \cos 2x$$

$$\Rightarrow \sin 2x = 0, 2 \cos^2 2x = 2 \cos^2 2x - 1 + \cos 2x$$

Trigonometric Equations

$$\Rightarrow \sin 2x = 0, \cos 2x = 1$$

$$\Rightarrow 2x = n\pi, 2x = 2n\pi$$

$$\Rightarrow x = \frac{n\pi}{2}, x = n\pi, n \in I$$

97. Given equation is

$$(1 - \tan x)(1 + \sin 2x) = (1 + \tan x)$$

$$\Rightarrow (1 - \tan x) \left(1 + \frac{2 \tan x}{1 + \tan^2 x} \right) = (1 + \tan x)$$

$$\Rightarrow (1 - \tan x)(1 + \tan x)^2 = (1 + \tan x)(1 + \tan^2 x)$$

$$\Rightarrow (1 - \tan^2 x)(1 + \tan x) = (1 + \tan x)(1 + \tan^2 x)$$

$$\Rightarrow ((1 - \tan^2 x) - (1 + \tan^2 x))(1 + \tan x) = 0$$

$$\Rightarrow \tan^2 x(1 + \tan x) = 0$$

$$\Rightarrow \tan^2 x = 0, (1 + \tan x) = 0$$

$$\Rightarrow \tan^2 x = 0, \tan x = 4$$

$$\Rightarrow x = n\pi, x = n\pi - \frac{\pi}{4}, n \in I$$

98. Given equation is

$$\sin x - 3 \sin 2x + \sin 3x = \cos x - 3 \cos 2x + \cos 3x$$

$$\Rightarrow (\sin 3x + \sin x) - 3 \sin 2x = (\cos 3x + \cos x) - 3 \cos 2x$$

$$\Rightarrow 2 \sin 2x \cos x - 3 \sin 2x = 2 \cos 2x \cos x - 3 \cos 2x$$

$$\Rightarrow \sin 2x(2 \cos x - 3) = (2 \cos x - 3) \cos 2x$$

$$\Rightarrow \frac{\sin 2x}{\cos 2x}(2 \cos x - 3) = (2 \cos x - 3)$$

$$\Rightarrow \frac{\sin 2x}{\cos 2x} = 1$$

$$\Rightarrow \tan 2x = 1$$

$$\Rightarrow 2x = n\pi + \frac{\pi}{4}$$

$$\Rightarrow x = \frac{n\pi}{2} + \frac{\pi}{8}, n \in I$$

99. The given equation can be written as

$$(2 \sin 2x \cdot \sin x) 2 \sin 4x - \sin 3x = 0$$

$$\Rightarrow 2(\cos x - \cos 3x) \sin 4x - \sin 3x = 0$$

$$\Rightarrow 2 \sin 4x \cos x - 2 \sin 4x \cos 3x - \sin 3x = 0$$

$$\Rightarrow (\sin 5x + \sin 3x) - (\sin 7x + \sin x) - \sin 3x = 0$$

$$\Rightarrow (\sin 7x - \sin 5x) + \sin x = 0$$

$$\Rightarrow \sin x(2 \cos 6x + 1) = 0$$

$$\Rightarrow \sin x = 0, \cos 6x = -1/2,$$

$$\Rightarrow \sin x = 0, \cos 6x = \cos \left(\frac{2\pi}{3} \right)$$

$$\Rightarrow x = n\pi, 6x = 2n\pi \pm \frac{2\pi}{3}, n \in Z$$

$$\Rightarrow x = n\pi, x = (3n \pm 1) \frac{\pi}{9}, n \in Z$$

100. Do yourself.

101. Do yourself.

102. Given equation is

$$\sin 4x \sin 2x = \cos 6x - \cos 2x$$

$$\Rightarrow \sin 4x \sin 2x = -2 \sin 4x \sin 2x$$

$$\Rightarrow 3 \sin 4x \sin 2x = 0$$

$$\Rightarrow \sin 4x = 0, \sin 2x = 0$$

$$\Rightarrow 4x = n\pi, 2x = n\pi, n \in I$$

$$\Rightarrow x = \frac{n\pi}{4}, x = \frac{n\pi}{2}, n \in I$$

103. Given equation is

$$\sec x \cos 5x + 1 = 0$$

$$\Rightarrow \cos 5x + \cos x = 0$$

$$\Rightarrow 2 \cos (3x) \cos (2x) = 0$$

$$\Rightarrow 2 \cos (3x) = 0, \cos (2x) = 0$$

$$\Rightarrow \cos (3x) = 0, \cos (2x) = 0$$

$$\Rightarrow 3x = (2n+1) \frac{\pi}{2}, 2x = (2n+1) \frac{\pi}{2}, n \in I$$

$$\Rightarrow x = (2n+1) \frac{\pi}{6}, x = (2n+1) \frac{\pi}{4}, n \in I$$

Hence, the solutions are

$$x = \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{5\pi}{6}$$

105. Given equation is

$$\cos (6x) \cos x = 1$$

$$\Rightarrow 2 \cos (6x) \cos x = -2$$

$$\Rightarrow \cos 7x + \cos 5x = -2$$

It is possible only when

$$\cos (7x) = -1, \cos (5x) = -1$$

$$\Rightarrow x = (2n+1) \frac{\pi}{7}, x = (2n+1) \frac{\pi}{5}, n \in I$$

106. The given equation can be written as

$$2 \sin^2 x - 5 \sin x \cos x - 8 \cos^2 x = -2(\sin^2 x + \cos^2 x)$$

$$\Rightarrow 2 \tan^2 x - 5 \tan x - 8 = -(\tan^2 x + 1)$$

$$\Rightarrow 4 \tan^2 x - 5 \tan x - 6 = 0$$

$$\Rightarrow (\tan x - 2)(4 \tan x + 3) = 0$$

$$\Rightarrow \tan x = -2, \tan x = -\frac{3}{4}$$

$$\Rightarrow x = n\pi + \alpha, x = n\pi + \beta, \text{ where}$$

$$\Rightarrow \alpha = \tan^{-1}(2), \beta = \tan^{-1} \left(-\frac{3}{4} \right), n \in Z.$$

107. Given equation is

$$5 \sin^2 x - 7 \sin x \cos x + 10 \cos^2 x = 4$$

$$\Rightarrow 5 \tan^2 x - 7 \tan x + 10 = 4 \sec^2 x$$

$$\Rightarrow 5 \tan^2 x - 7 \tan x + 10 = 4 + 4 \tan^2 x$$

$$\Rightarrow \tan^2 x - 7 \tan x + 6 = 0$$

$$\Rightarrow (\tan x - 1)(\tan x - 6) = 0$$

$$\Rightarrow \tan x = 1, 6$$

$$\Rightarrow x = n\pi + \frac{\pi}{4}, x = n\pi + \alpha, \alpha = \tan^{-1}(5)$$

108. Given equation is

$$\begin{aligned} 2 \sin^2 x - 5 \sin x \cos x - 8 \cos^2 x &= -3 \\ \Rightarrow 2 \tan^2 x - 5 \tan x - 8 &= -3 \sec^2 x \\ \Rightarrow 2 \tan^2 x - 5 \tan x - 8 &= -3 - 3 \tan^2 x \\ \Rightarrow 5 \tan^2 x - 5 \tan x - 5 &= 0 \\ \Rightarrow \tan^2 x - \tan x - 1 &= 0 \end{aligned}$$

$$\Rightarrow \tan x = \frac{1 \pm \sqrt{5}}{2}$$

$$\Rightarrow x = n\pi + \alpha, \alpha = \tan^{-1} \left(\frac{1 \pm \sqrt{5}}{2} \right)$$

109. Given equation is

$$\begin{aligned} \sin^2 x \cos x + \sin^2 x \cos^2 x + \sin x \cos^3 x &= 1 \\ \Rightarrow \sin x \cos x [\sin^2 x + \sin x \cos x + \cos^2 x] &= 1 \\ \Rightarrow \sin x \cos x [1 + \sin x \cos x] &= 1 \\ \Rightarrow 2 \sin x \cos x [2 + 2 \sin x \cos x] &= 4 \\ \Rightarrow \sin(2x) (2 + \sin(2x)) &= 4 \\ \Rightarrow \sin^2(2x) + 2 \sin(2x) - 4 &= 0 \end{aligned}$$

$$\Rightarrow \sin(2x) = \frac{-2 \pm \sqrt{20}}{2}$$

$$\Rightarrow \sin(2x) = \frac{-2 \pm 2\sqrt{5}}{2} = -1 \pm \sqrt{5}$$

It is not possible.

So, it has no solution.

110. The given equation can be written as

$$\begin{aligned} 1 + \frac{2}{\sin x} &= -\frac{1}{2} \left(1 + \tan^2 \left(\frac{x}{2} \right) \right) \\ \Rightarrow 2(\sin x + 2) &= - \left(1 + \tan^2 \left(\frac{x}{2} \right) \right) \sin x \\ \Rightarrow 2 \left(\frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} + 2 \right) &= - \left(1 + \tan^2 \frac{x}{2} \right) \cdot \left(\frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \right) \\ \Rightarrow 2 \left(\frac{2t}{1+t^2} + 2 \right) &= -(1+t^2) \times \left(\frac{2t}{1+t^2} \right), \end{aligned}$$

 where $t = \tan(x/2)$

$$\begin{aligned} \Rightarrow t^3 + 2t^2 + 3t + 2 &= 0 \\ \Rightarrow t^3 + t^2 + t^2 + t + 2t + 2 &= 0 \\ \Rightarrow (t+1)(t^2 + t + 2) &= 0 \\ \Rightarrow t+1=0, t^2+t+2 \neq 0 \end{aligned}$$

$$\Rightarrow \tan \left(\frac{x}{2} \right) = -1 = \tan \left(\frac{-\pi}{4} \right)$$

$$\Rightarrow \frac{x}{2} = n\pi - \frac{\pi}{4}, n \in \mathbb{Z}$$

$$\Rightarrow x = 2n\pi - \frac{\pi}{2}, n \in \mathbb{Z}.$$

111. Given equation is

$$\begin{aligned} (\cos x - \sin x)(2 \tan x + \sec x) + 2 &= 0 \\ \Rightarrow (\cos x - \sin x)(2 \sin x + 1) + 2 \cos x &= 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow \left(\frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)} - \frac{2 \tan(x/2)}{1 + \tan^2(x/2)} \right) \left(\frac{4 \tan(x/2)}{1 + \tan^2(x/2)} + 1 \right) \\ + 2 \left(\frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)} \right) &= 0 \end{aligned}$$

 Put $\tan \left(\frac{x}{2} \right) = t$ and then solve it.

112. Given equation is

$$\begin{aligned} \frac{\sin^3 \frac{x}{2} - \cos^3 \frac{x}{2}}{2 + \sin x} &= \frac{\cos x}{3} \\ \Rightarrow \frac{\left(\sin \left(\frac{x}{2} \right) - \cos \left(\frac{x}{2} \right) \right) \left(1 + \frac{\sin x}{2} \right)}{2 + \sin x} \\ &= \frac{\left(\cos^2 \left(\frac{x}{2} \right) - \sin^2 \left(\frac{x}{2} \right) \right)}{3} \\ \Rightarrow -\frac{3}{2} &= \left(\cos \left(\frac{x}{2} \right) + \sin \frac{x}{2} \right) \end{aligned}$$

It is not possible.

So, it has no solution.

113. Given equation is

$$\begin{aligned} \Rightarrow \cot \left(\frac{x}{2} \right) - \operatorname{cosec} \left(\frac{x}{2} \right) &= \cot x \\ \frac{\cos \left(\frac{x}{2} \right) - 1}{\sin \left(\frac{x}{2} \right)} &= \cot x \\ \Rightarrow 2 \sin^2 \left(\frac{x}{2} \right) &= -\sin \left(\frac{x}{2} \right) \cot x \\ \Rightarrow \left(2 \sin \left(\frac{x}{2} \right) + \cot x \right) \sin \left(\frac{x}{2} \right) &= 0 \\ \Rightarrow \left(2 \sin \left(\frac{x}{2} \right) + \cot x \right) &= 0, \sin \left(\frac{x}{2} \right) = 0 \end{aligned}$$

 when $\sin \left(\frac{x}{2} \right) = 0$

$$\Rightarrow x = 2n\pi, n \in \mathbb{Z}$$

 when $\left(2 \sin \left(\frac{x}{2} \right) + \cot x \right) = 0$

$$\Rightarrow 2 \sin \left(\frac{x}{2} \right) = -\frac{\cos x}{\sin x}$$

$$\Rightarrow 2 \sin \left(\frac{x}{2} \right) = -\frac{\left(\cos^2 \left(\frac{x}{2} \right) - \sin^2 \left(\frac{x}{2} \right) \right)}{2 \sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right)}$$

$$\Rightarrow 4 \sin^2\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right) + \cos^2\left(\frac{x}{2}\right) - \sin^2\left(\frac{x}{2}\right) = 0$$

$$\Rightarrow \cos^2\left(\frac{x}{2}\right) + 4 \sin^2\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right) - \sin^2\left(\frac{x}{2}\right) = 0$$

For all real x ,

$$16 \sin^4\left(\frac{x}{2}\right) + 4 \sin^2\left(\frac{x}{2}\right) \geq 0$$

$$\Rightarrow 4 \sin^4\left(\frac{x}{2}\right) + \sin^2\left(\frac{x}{2}\right) = 0$$

$$\Rightarrow \sin^2\left(\frac{x}{2}\right) \left(4 \sin^2\left(\frac{x}{2}\right) + 1 \right) = 0$$

$$\Rightarrow \sin^2\left(\frac{x}{2}\right) = 0, \left(4 \sin^2\left(\frac{x}{2}\right) + 1 \right) = 0$$

$$\Rightarrow \sin^2\left(\frac{x}{2}\right) = 0$$

$$\Rightarrow \sin\left(\frac{x}{2}\right) = 0$$

$$\Rightarrow x = 2n\pi, n \in I$$

114. Given equation is

$$\sin(\theta + \alpha) = k \sin(2\theta)$$

$$\Rightarrow \sin \theta \cos \alpha + \cos \theta \sin \alpha = k \sin(2\theta)$$

$$\Rightarrow \left(\frac{2 \tan\left(\frac{\theta}{2}\right)}{1 + \tan^2\left(\frac{\theta}{2}\right)} \right) \cos \alpha + \left(\frac{1 - \tan^2\left(\frac{\theta}{2}\right)}{1 + \tan^2\left(\frac{\theta}{2}\right)} \right) \sin \alpha$$

$$= 2k \left(\frac{2 \tan\left(\frac{\theta}{2}\right)}{1 + \tan^2\left(\frac{\theta}{2}\right)} \right) \left(\frac{1 - \tan^2\left(\frac{\theta}{2}\right)}{1 + \tan^2\left(\frac{\theta}{2}\right)} \right)$$

$$\Rightarrow \left(\frac{2t}{1+t^2} \right) \cos \alpha + \left(\frac{1-t^2}{1+t^2} \right) \sin \alpha$$

$$= 2k \cdot \left(\frac{2t}{1+t^2} \right) \left(\frac{1-t^2}{1+t^2} \right)$$

$$\Rightarrow 2t(1+t^2) \cos \alpha + (1-t^4) \sin \alpha = 4k t(1-t^2)$$

$$\Rightarrow (\sin \alpha) t^4 - (4k + 2 \cos \alpha) t^3 + (4k - 2 \cos \alpha) t = \sin \alpha = 0$$

Let t_1, t_2, t_3 and t_4 be its four roots

$$\Sigma t_1 = \frac{4k + 2 \cos \alpha}{\sin \alpha} = s_1$$

$$\Sigma t_1 t_2 = 0 = s_2$$

$$\Sigma t_1 t_2 t_3 = \frac{2 \cos \alpha - 4k}{\sin \alpha} = s_3$$

$$\Sigma t_1 t_2 t_3 t_4 = -\frac{\sin \alpha}{\sin \alpha} = -1 = s_4$$

$$\text{Now, } \tan\left(\frac{\theta_1 + \theta_2 + \theta_3 + \theta_4}{2}\right) = \frac{s_1 - s_3}{1 - s_2 + s_4}$$

$$\Rightarrow \tan\left(\frac{\theta_1 + \theta_2 + \theta_3 + \theta_4}{2}\right) = \infty$$

$$\Rightarrow \tan\left(\frac{\theta_1 + \theta_2 + \theta_3 + \theta_4}{2}\right) = \tan\left(\frac{\pi}{2}\right)$$

$$\Rightarrow \left(\frac{\theta_1 + \theta_2 + \theta_3 + \theta_4}{2}\right) = n\pi + \left(\frac{\pi}{2}\right)$$

$$\Rightarrow (\theta_1 + \theta_2 + \theta_3 + \theta_4) = (2n+1)\pi, n \in I$$

115. Let $\sin x + \cos x = t$

$$\Rightarrow \sin x \cdot \cos x = \frac{t^2 - 1}{2}$$

So, the given equation can be reduces to

$$t - 2\sqrt{2} \left(\frac{t^2 - 1}{2} \right) = 0$$

$$\Rightarrow \sqrt{2} t^2 - t - \sqrt{2} = 0$$

$$\Rightarrow (\sqrt{2}t + 1)(t - \sqrt{2}) = 0$$

$$\Rightarrow t = \sqrt{2}, -\frac{1}{\sqrt{2}}$$

When $\sin x + \cos x = \sqrt{2}$

$$\Rightarrow \frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x = 1$$

$$\Rightarrow \sin\left(x + \frac{\pi}{4}\right) = 1 = \sin\left(\frac{\pi}{2}\right)$$

$$\Rightarrow x + \frac{n\pi}{4} = n\pi + (-1)^n \frac{\pi}{2}, n \in Z$$

When $\sin x + \cos x = -\frac{1}{\sqrt{2}}$

$$\Rightarrow \frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x = -\frac{1}{2}$$

$$\Rightarrow \sin\left(x + \frac{\pi}{4}\right) = \sin\left(-\frac{\pi}{6}\right)$$

$$\Rightarrow \left(x + \frac{\pi}{4}\right) = n\pi + (-1)^n \left(-\frac{\pi}{6}\right)$$

$$\Rightarrow x = n\pi - (-1)^n \frac{\pi}{6} - \frac{\pi}{4}, n \in Z$$

117. Given equation is

$$\sin x + \cos x = 1 - \sin x \cos x$$

...(i)

Put $\sin x + \cos x = t$

$$\Rightarrow \sin x \cdot \cos x = \frac{t^2 - 1}{2}$$

Now, equation (i) becomes

$$t = 1 - \frac{t^2 - 1}{2}$$

$$\begin{aligned} \Rightarrow 2t &= 2 - t^2 + 1 \\ \Rightarrow t^2 + 2t - 3 &= 0 \\ \Rightarrow (t+3)(t-1) &= 0 \\ \Rightarrow (t+3) &= 0, (t-1) = 0 \\ \Rightarrow \sin x + \cos x &= 1, \sin x + \cos x = -3 \\ \Rightarrow \sin x + \cos x &= 1 \end{aligned}$$

$$\Rightarrow \frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \sin\left(x + \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$\Rightarrow x = n\pi + (-1)^n \left(\frac{\pi}{4}\right) - \frac{\pi}{4}, n \in I$$

118. Given equation is

$$1 + \sin^3 x + \cos^3 x = \frac{3}{2} \sin 2x \quad \dots(i)$$

$$\Rightarrow 1 + (\sin x + \cos x)(1 - \sin x \cos x) = 3 \sin x \cos x$$

Put $\sin x + \cos x = t$

$$\Rightarrow \sin x \cdot \cos x = \frac{t^2 - 1}{2}$$

Now, equation (i) becomes

$$1 + t \left(1 - \frac{t^2 - 1}{2}\right) = \frac{3}{2}(t^2 - 1)$$

$$\begin{aligned} \Rightarrow 2 + t(3 - t^2) &= 3(t^2 - 1) \\ \Rightarrow 2 + 3t - t^3 - 3t^2 + 3 &= 0 \\ \Rightarrow 3t - t^3 - 3t^2 + 5 &= 0 \\ \Rightarrow t^3 + 3t^2 - 3t - 5 &= 0 \\ \Rightarrow t^3 + t^2 + 2t^2 + 2t - 5t - 5 &= 0 \\ \Rightarrow t^2(t+1) + 2t(t+1) - 5(t+1) &= 0 \\ \Rightarrow (t^2 + 2t - 5)(t+1) &= 0 \end{aligned}$$

$$\Rightarrow t = -1, t = \frac{-2 \pm \sqrt{24}}{2}$$

$$\Rightarrow \sin x + \cos x = -1, \frac{-2 \pm \sqrt{24}}{2}$$

$$\Rightarrow \sin x + \cos x = -1$$

$$\Rightarrow \frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x = -\frac{1}{\sqrt{2}}$$

$$\Rightarrow \sin\left(x + \frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$$

$$\Rightarrow x = n\pi + (-1)^n \left(-\frac{\pi}{4}\right) - \frac{\pi}{4}, n \in I$$

119. Given equation is

$$\sin 2x - 12(\sin x - \cos x) + 12 = 0$$

$$2 \sin x \cos x - 12(\sin x - \cos x) + 12 = 0$$

$$\sin x \cos x - 6(\sin x - \cos x) + 6 = 0 \quad \dots(i)$$

Put $\sin x + \cos x = t$

$$\Rightarrow \sin x \cdot \cos x = \frac{t^2 - 1}{2}$$

Now, equation (i) becomes

$$\Rightarrow \frac{t^2 - 1}{2} - 6t + 6 = 0$$

$$\Rightarrow t^2 - 1 - 12t + 12 = 0$$

$$\Rightarrow t^2 - 12t + 11 = 0$$

$$\Rightarrow (t-1)(t-11) = 0$$

$$\Rightarrow t = 1, 11$$

$$\Rightarrow \sin x + \cos x = 1$$

$$\Rightarrow \sin\left(x + \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$\Rightarrow x = n\pi + (-1)^n \left(\frac{\pi}{4}\right) - \frac{\pi}{4}, n \in I$$

Hence, the solution are

$$x = 0, \frac{\pi}{2}, 2\pi$$

120. The given equation can be written as

$$\sin 6x + \cos 4x = -2$$

$$\Rightarrow \sin 6x = -1 \text{ and } \cos 4x = -1$$

$$\Rightarrow \sin 6x = \sin \frac{3\pi}{2}, \cos 4x = \cos \pi$$

$$\Rightarrow 6x = 2n\pi + \frac{3\pi}{2}, 4x = 2n\pi + \pi, n \in Z$$

$$\Rightarrow x = \frac{n\pi}{3} + \frac{\pi}{4}, x = \frac{n\pi}{2} + \frac{\pi}{4}, n \in Z$$

$$\Rightarrow x = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{12}, \frac{11\pi}{12}, \dots$$

$$\dots x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \dots$$

$$\Rightarrow x = \frac{\pi}{4}, \frac{5\pi}{4}$$

Hence, the general solution will be,

$$\Rightarrow x = 2n\pi + \frac{\pi}{4}, 2n\pi + \frac{5\pi}{4}, n \in Z$$

$$\Rightarrow x = 2n\pi + \frac{\pi}{4}, (2n+1)\pi + \frac{\pi}{4}, n \in Z$$

$$\Rightarrow x = m\pi + \frac{\pi}{4}, m \in Z$$

122. Given equation is

$$\sin^4 x = 1 + \tan^8 x$$

It is possible only when

$$\sin^4 x = 1, \tan^8 x = 0$$

$$\Rightarrow \sin^2 x = 1, \tan x = 0$$

$$\Rightarrow x = n\pi \pm \frac{\pi}{2}, x = n\pi, n \in I$$

There is no common value which satisfies both the above equations.

Hence, the equation has no solution.

123. Given $\sin^2 x + \cos^2 y = 2 \sec^2 z$

Here, $\text{LHS} \leq 2$ and $\text{RHS} \geq 2$

It is possible only when

$$\sin^2 x = 1, \cos^2 y = 1, \sec^2 z = 1$$

$$\Rightarrow \cos^2 x = 0, \sin^2 y = 0, \cos^2 z = 1$$

$$\Rightarrow \cos^2 x = 0, \sin^2 y = 0, \sin^2 z = 1$$

$$\Rightarrow \cos x = 0, \sin y = 0, \sin z = 0$$

$$\Rightarrow x = (2n+1)\frac{\pi}{2}, y = m\pi, z = k\pi$$

where, $n, m, k \in I$

124. Given equation is

$$\sin 3x + \cos 2x + 2 = 0$$

It is possible only when

$$\sin 3x = -1, \cos 2x = -1$$

$$\Rightarrow 3x = \frac{3\pi}{2}, 2x = \pi$$

$$\Rightarrow x = \frac{\pi}{2}, x = \frac{\pi}{2}$$

Hence, the general solution is

$$x = 2n\pi + \frac{\pi}{2}, n \in I$$

125. Given equation is $\cos 4x + \sin 5x = 2$

It is possible only when

$$\cos 4x = 1, \sin 5x = 1$$

$$\Rightarrow 4x = 2n\pi, 5x = (4n+1)\frac{\pi}{2}$$

$$\Rightarrow x = \frac{2n\pi}{4}, x = (4n+1)\frac{\pi}{10}$$

Thus, $x = \frac{\pi}{2}$ satisfies both

Hence, the solution is

$$x = 2n\pi + \frac{\pi}{2} = (4n+1)\frac{\pi}{2}, n \in I$$

126. The given equation can be written as

$$8^{1+|\cos x|+\cos^2 x+|\cos x|^3+\cos^4 x+|\cos x|^5+\dots\text{to } \infty} = 8^2$$

$$\Rightarrow 1 + |\cos x| + \cos^2 x + |\cos x|^3 + \cos^4 x + |\cos x|^5 + \dots \text{to } \infty = 2$$

$$\Rightarrow \frac{1}{1-|\cos x|} = 2$$

$$\Rightarrow \cos x = \pm \frac{1}{2}$$

When $\cos x = \frac{1}{2}$

$$\Rightarrow \cos x = \cos\left(\frac{\pi}{3}\right)$$

$$\Rightarrow x = 2n\pi \pm \frac{\pi}{3}, n \in Z$$

When $\cos x = -\frac{1}{2} = \cos\left(\frac{2\pi}{3}\right)$

$$\Rightarrow x = 2n\pi \pm \frac{2\pi}{3}, n \in Z$$

Hence the values of x are $\pm \frac{\pi}{3}, \pm \frac{2\pi}{3}$.

127. Given equation is

$$2^{1+|\cos x|+\cos^2 x+|\cos x|^3+\cos^4 x+|\cos x|^5+\dots\text{to } \infty} = 4$$

$$\Rightarrow 2^{\frac{1}{1-|\cos x|}} = 4 = 2^2$$

$$\Rightarrow \frac{1}{1-|\cos x|} = 2$$

$$\Rightarrow 1-|\cos x| = \frac{1}{2}$$

$$\Rightarrow |\cos x| = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\Rightarrow \cos x = \pm \frac{1}{2}$$

Hence the values of x are $\pm \frac{\pi}{3}, \pm \frac{2\pi}{3}$.

128. Given equation is

$$1 + \sin \theta + \sin^2 \theta + \dots = 4 + 2\sqrt{3}$$

$$\Rightarrow \frac{1}{1-\sin \theta} = 4 + 2\sqrt{3}$$

$$\Rightarrow 1-\sin \theta = \frac{1}{4+2\sqrt{3}}$$

$$\Rightarrow \sin \theta = 1 - \frac{1}{4+2\sqrt{3}}$$

$$\Rightarrow \sin \theta = 1 - \frac{1}{4+2\sqrt{3}} = 1 - \frac{2-\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \theta = n\pi + (-1)^n \left(\frac{\pi}{3}\right), n \in I$$

129. Given equation is

$$|\cos x|^{\sin^2 x - \frac{3}{2}\sin x + \frac{1}{2}} = 1$$

$$\Rightarrow \left(\sin^2 x - \frac{3}{2}\sin x + \frac{1}{2}\right) \log |\cos x| = 0$$

$$\Rightarrow (2\sin^2 x - 3\sin x + 1) \log |\cos x| = 0$$

$$\Rightarrow (\sin x - 1)(2\sin x - 1) \log |\cos x| = 0$$

$$\begin{aligned} \Rightarrow (\sin x - 1) &= 0, (2 \sin x - 1) = 0, \log |\cos x| = 0 \\ \Rightarrow \sin x &= 1, \sin x = \frac{1}{2}, \log |\cos x| = 0 \\ \Rightarrow \sin x &= \frac{1}{2}, |\cos x| = 1 \\ \Rightarrow \sin x &= \frac{1}{2}, \cos = 1, \cos x = -1 \\ \Rightarrow x &= n\pi + (-1)^n \left(\frac{\pi}{6} \right), x = 2n\pi, x = (2n+1)\pi \end{aligned}$$

130. Given equation is

$$\begin{aligned} e^{\sin x} - e^{-\sin x} - 4 &= 0 \\ \Rightarrow t - \frac{1}{t} - 4 &= 0, t = e^{\sin x} \\ \Rightarrow t^2 - 4t - 1 &= 0 \\ \Rightarrow (t-2)^2 &= 5 \\ \Rightarrow t &= 2 \pm \sqrt{5} \\ \Rightarrow e^{\sin x} &= 2 \pm \sqrt{5} \\ \Rightarrow \sin x &= \log_e (2 \pm \sqrt{5}) \\ \Rightarrow \sin x &= \log_e (2 + \sqrt{5}) \\ \Rightarrow \sin x &= \log_e (2 + \sqrt{5}) > 1 \end{aligned}$$

It is not possible

So, it has no solution.

131. We have

$$\begin{aligned} e^{[\sin^2 x + \sin^4 x + \sin^6 x + \dots \text{to } \infty] \log_e 2} \\ = e^{\left(\frac{\sin^2 x}{1 - \cos^2 x} \right) \log_e 2} = e^{\tan^2 x \log_e 2} = 2^{\tan^2 x} \end{aligned}$$

$$\text{Let } a = 2^{\tan^2 x}$$

$$\text{Thus, } a^2 - 9a + 8 = 0$$

$$\Rightarrow (a-1)(a-8) = 0$$

$$\Rightarrow a = 1, 8$$

$$\text{when } a = 1, \text{ then } 2^{\tan^2 x} = 1 = 2^0$$

$$\Rightarrow 2^{\tan^2 x} = 1 = 2^0$$

$$\Rightarrow \tan^2 x = 0$$

$$\Rightarrow x = n\pi, n \in I$$

$$\text{when } a = 8, \text{ then } 2^{\tan^2 x} = 8 = 2^3$$

$$\Rightarrow \tan^2 x = 3$$

$$\Rightarrow \tan x = \sqrt{3}$$

$$\text{Now, } \frac{\cos x}{\cos x + \sin x}$$

$$= \frac{1}{1 + \tan x} = \frac{1}{\sqrt{3} + 1}$$

$$= \frac{(\sqrt{3} - 1)}{2}$$

132. Given equation is

$$\log_{\cos x} \tan x + \log_{\sin x} \cot x = 0$$

$$\Rightarrow \log_{\cos x} \left(\frac{\sin x}{\cos x} \right) + \log_{\sin x} \left(\frac{\cos x}{\sin x} \right) = 0$$

$$\Rightarrow \log_{\cos x} (\sin x) + \log_{\sin x} (\cos x) = 2$$

It is possible only when

$$\sin x = \cos x$$

$$\Rightarrow \tan x = 1$$

$$\Rightarrow x = \frac{\pi}{4}$$

133. Given equation is

$$3^{\sin 2x + 2 \cos^2 x} + 3^{1 - \sin 2x + 2 \sin^2 x} = 28$$

$$\Rightarrow 3^{\sin 2x + 2 \cos^2 x} + 3^{1 - \sin 2x + 2 - 2 \cos^2 x} = 28$$

$$\Rightarrow 3^{\sin 2x + 2 \cos^2 x} + \frac{3^3}{3^{\sin 2x + 2 \cos^2 x}} = 28$$

$$\Rightarrow a + \frac{27}{a} = 28, a = 3^{\sin 2x + 2 \cos^2 x}$$

$$\Rightarrow a^2 - 28a + 27 = 0$$

$$\Rightarrow (a-27)(a-1) = 0$$

$$\Rightarrow a = 27, 1$$

$$\text{when } a = 27, \text{ then } 3^{\sin 2x + 2 \cos^2 x} = 3^3$$

$$\Rightarrow \sin 2x + 2 \cos^2 x = 3$$

$$\Rightarrow \sin 2x + 1 + \cos 2x = 3$$

$$\Rightarrow \sin 2x + \cos 2x = 2$$

It is not possible.

$$\text{when } a = 1, \text{ then } 3^{\sin 2x + 2 \cos^2 x} = 3^0$$

$$\Rightarrow \sin 2x + 2 \cos^2 x = 0$$

$$\Rightarrow \sin 2x + 1 + \cos 2x = 0$$

$$\Rightarrow \sin 2x + \cos 2x = -1$$

$$\Rightarrow \frac{1}{\sqrt{2}} \sin 2x + \frac{1}{\sqrt{2}} \cos 2x = -\frac{1}{\sqrt{2}}$$

$$\Rightarrow \sin \left(2x + \frac{\pi}{4} \right) = -\frac{1}{\sqrt{2}}$$

$$\Rightarrow \left(2x + \frac{\pi}{4} \right) = n\pi + (-1)^n \left(-\frac{\pi}{4} \right)$$

$$\Rightarrow x = \frac{n\pi}{2} + (-1)^n \left(-\frac{\pi}{8} \right) - \frac{\pi}{8}, n \in I$$

134. Do yourself.

135. Given equation is

$$2 \cos^2 \left(\frac{x}{2} \right) \sin^2 x = x^2 + \frac{1}{x^2}$$

$$\text{Here, LHS} < 2 \text{ for } 0 < x < \frac{\pi}{2}$$

$$\text{and RHS} \geq 2$$

So, it has no solutions.